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A constrained variational method for crack networks with criterion-based evolution

Nasr Ghoniem^{1*}

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*Correspondence:

Nasr Ghoniem
ghoniem@ucla.edu

¹Mechanical & Aerospace
Engineering Department,
University of California at Los
Angeles, 420 Westwood Plaza, Los
Angeles, California 90095, USA

Abstract

A constrained Variational Crack Element Method (VCEM) for interacting crack networks in two-dimensional isotropic elasticity is formulated. The method is variational at the level of crack equilibrium for fixed geometry, while crack advance is treated separately using near-tip propagation criteria. Cracks are represented by distributed Burgers-density fields, preserving the exact crack-tip singular structure within a unified energy-minimization framework. The governing equations follow from a constrained variational problem in which equality constraints enforce Burgers-vector closure and displacement single-valuedness at junctions, while inequality constraints enforce crack-face non-penetration. The resulting system has a Karush–Kuhn–Tucker (KKT) saddle-point structure. Crack geometry is represented by graph-based polylines composed of polarity-carrying half-crack elements, allowing the treatment of tips, kinks, branching, and multi-degree junctions. Elastostatic interactions are expressed through tensor kernels derived from Kelvin's fundamental solutions, and both strong (collocation) and weak (Galerkin/least-squares) traction enforcement are implemented. The framework provides a sharp-interface, computationally scalable approach for equilibrium and criterion-based evolution of complex crack networks in finite or infinite domains.

Keywords Variational, Fracture, Crack networks, Crack propagation

Introduction

The computational modeling of interacting crack networks remains challenging because the problem is simultaneously nonlocal, singular, and topologically complex. Each crack perturbs the stress field of all others through long-range elastic interactions, while crack growth introduces evolving tips, kinks, branching events, and junctions that must be handled without loss of near-tip fidelity. These difficulties are amplified by the wide separation of scales between crack-tip fields and network-scale response, and by the need to evaluate local propagation criteria repeatedly as the network evolves (Lawn 1993; Movchan et al. 1998).

Existing computational approaches address different parts of this difficulty, but none resolves all of them efficiently. Classical FEM is versatile but generally requires local mesh refinement, singular elements, or remeshing as cracks propagate (Zienkiewicz and

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Taylor 2005; Barsoum 1976; Henshell and Shaw 1975; Bouchard et al. 2003). XFEM alleviates mesh-alignment constraints, but the enrichment structure becomes increasingly complicated and often ill-conditioned for multiple interacting cracks (Moës et al. 1999; Belytschko et al. 2001; Sukumar et al. 2000; Karihaloo and Xiao 2003; Béchet et al. 2005; Menk and Bordas 2011). BEM and distributed-dislocation methods preserve the sharp crack singularity and naturally encode long-range elastic interactions, but they lead to dense operators and become difficult to manage when crack topology evolves (Crouch and Starfield 1983; Bonnet 1999; Crouch 1976; Pan 1997; Zhang and Gross 2002). Phase-field methods handle branching, merging, and nucleation naturally, but at the cost of a diffuse crack representation, volumetric discretization, and a regularization length scale that can compromise sharp-crack accuracy in densely interacting networks (Bourdin et al. 2008; Francfort and Marigo 1998; Miehe et al. 2010; Borden et al. 2012; Bourdin et al. 2014; Mikelić et al. 2015; Linse et al. 2017; Tanné et al. 2018; Alabdullah and Ghoniem 2024).

The objective of the present work is to develop a formulation that retains the sharp-interface fidelity of dislocation-based crack mechanics while providing a systematic framework for crack-network topology, constraint enforcement, and scalable computation. To this end, we introduce a Variational Crack Element Method (VCEM) in which cracks are represented by distributed Burgers-density fields defined on graph-structured crack networks. This representation preserves the singular crack-tip structure of linear elastic fracture mechanics while allowing crack systems to be described in terms of tips, kinks, branches, and junctions.

The framework developed here is variational in a restricted but precise sense: for a fixed crack geometry, the crack-field degrees of freedom are determined by the constrained minimization of the elastic energy. The resulting stationarity conditions yield a Karush–Kuhn–Tucker (KKT) saddle-point system in which equality constraints enforce junction compatibility and Burgers-vector closure, while inequality constraints enforce crack-face non-penetration. Crack evolution, however, is not posed here as a global or incremental minimization over admissible crack geometries. Rather, once equilibrium has been established for a given configuration, crack advance is introduced separately through local propagation criteria. Thus, the variational character of the present formulation pertains to the equilibrium subproblem, not to a full Francfort–Marigo-type evolution of fracture (Francfort and Marigo 1998). The present formulation is fundamentally two-dimensional. The crack representation, interaction kernels, near-tip field extraction, and numerical implementation are all developed for planar crack networks in 2D linear elasticity. Extension to fully three-dimensional fracture would require substantial additional developments, including a surface-based representation of displacement discontinuities, treatment of crack fronts rather than crack tips, and a reformulation of the associated kernels and topological update rules. The current work should therefore be understood as a 2D formulation and validation study, intended as a foundation for future extensions. A schematic overview of the method is shown in Fig. 1.

The principal contributions of this paper are fourfold. First, crack-network equilibrium for fixed geometry is formulated as a constrained variational problem in terms of distributed Burgers-density degrees of freedom. Second, the resulting stationarity conditions lead naturally to a KKT structure that accommodates junction compatibility, Burgers-vector closure, and unilateral crack-face constraints within a unified framework. Third,

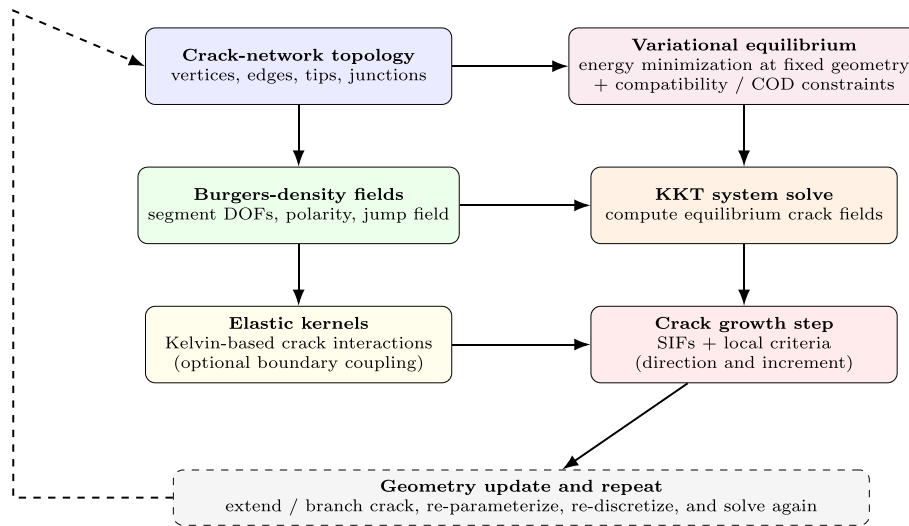


Fig. 1 Schematic overview of the Variational Crack Element Method (VCEM). The framework combines crack-network topology, Burgers-density crack representation, kernel-based elastic interactions, constrained variational equilibrium solved through a KKT system, and criterion-based crack growth with geometric update

crack topology is represented through a graph-based structure that serves as the computational backbone for classifying tips, kinks, and multi-branch junctions, and for assembling the corresponding compatibility and closure constraints. Fourth, the formulation is extended to bounded domains through coupling with a boundary-element representation of the outer boundary. Together, these ingredients define a computational framework for interacting crack networks that preserves sharp crack kinematics while remaining compatible with large-scale constrained solution strategies.

The paper is organized as follows. [Crack network topology](#) section introduces the graph-based representation of crack geometry and connectivity. [Displacement and stress fields](#) section develops the displacement and stress fields generated by distributed Burgers densities. [Saddle point constrained variational formulation](#) section presents the constrained variational formulation, its KKT structure, and its coupling to finite-domain boundary representations. [Crack network evolution](#) section describes the criterion-based crack-growth framework, including stress-intensity evaluation and propagation rules. Finally, [Conclusions and outlook](#) section summarizes the main conclusions and future directions.

Crack network topology

We represent a crack network as a finite undirected graph $G = (V, E)$ together with a geometric embedding $\mathcal{G} : G \rightarrow \mathbb{R}^2$. Here, $V = \{v_i\}_{i=1}^{N_v}$ denotes the set of vertices and $E = \{e_j\}_{j=1}^{N_e}$ the set of edges. Each edge $e = (v_i, v_k) \in E$ connects two distinct vertices $v_i \neq v_k$, and its geometric support is a parametric curve $\Gamma_e = \mathcal{G}(e)$ in the physical domain. Thus, the crack set is given by $\Gamma = \bigcup_{e \in E} \Gamma_e$. The topology of the network is encoded through the vertex–edge incidence map $\partial : E \rightarrow V \times V$, or equivalently by an incidence matrix $B \in \mathbb{R}^{N_v \times N_e}$ with entries $B_{ie} = \pm 1$ if vertex v_i is an endpoint of edge e , and 0 otherwise (with orientation assigned arbitrarily). This representation separates topology from geometry: G defines connectivity, while Γ_e defines geometric realization. The polyline geometry is shown in Fig. 2.

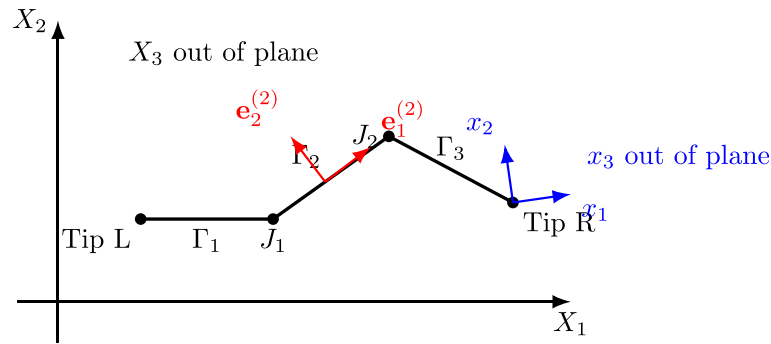


Fig. 2 Polyline crack as a graph of straight segments Γ_e connected at junctions J_k . Each segment carries a local frame (x_1, x_2, x_3)

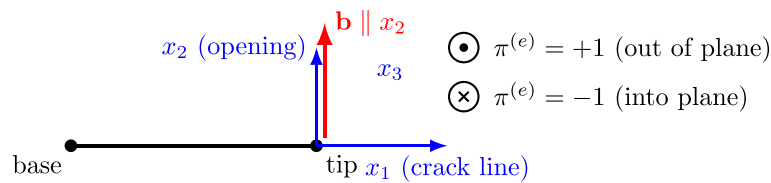


Fig. 3 Polar crack element with local axes. Mode-I Burgers vector is aligned with x_2 . Polarity is defined by the right-hand rule on x_3

Graph-based descriptions of crack or defect networks are, of course, not new in themselves and are used in several related fields. In particular, the vertex degree distinguishes tips, kinks, and multi-branch junctions; edge orientation and polarity define consistent jump conventions; and the resulting incidence structure determines how compatibility, Burgers-vector closure, and inequality constraints are assembled within the constrained variational/KKT formulation. Thus, the graph description is not a separate graph-theoretic contribution, but the organizational framework through which the mechanical constraint structure of the crack network is represented computationally.

The degree of a vertex $v \in V$ is defined as $\deg(v) = \#\{e \in E : v \in \partial e\}$, i.e., the number of incident edges. Vertices with $\deg(v) = 1$ correspond to crack tips, $\deg(v) = 2$ to interior kinks (curvature discontinuities), and $\deg(v) \geq 3$ to junctions where multiple branches interact. This classification provides a natural partition of the crack set into tip-dominated, kink-controlled, and junction-controlled regions, which subsequently governs the structure of the variational constraints imposed on the distributed Burgers densities. If the polyline crack has only two $\deg(V) = 1$ vertices, it will be designated as a *dipolar* (full) crack since the contour integral of the displacement vector around the crack is zero. If it has only one $\deg(V) = 1$ vertex, then it is designated as a *polar* (half) crack. This, for example, can represent cracks that intersect free surfaces or end at junctions ($\deg(v) \geq 3$).

Let (X_1, X_2) be the global coordinates in the plane, and let X_3 be the out-of-plane axis. Each segment e has a local orthonormal frame $(e_1^{(e)}, e_2^{(e)}, e_3^{(e)})$: $e_1^{(e)}$ = unit tangent along the segment, $e_2^{(e)}$ = unit normal to the crack face, $e_3^{(e)} = e_1^{(e)} \times e_2^{(e)}$. The local coordinates are $(x_1^{(e)}, x_2^{(e)}, x_3^{(e)})$, with $x_3^{(e)} \equiv X_3$.

When cracks terminate at free surfaces or internal junctions, they acquire polarity, as their net Burgers content does not vanish. Define element polarity $\pi^{(e)} \in \{+1, -1\}$ by the right-hand rule, as shown in Fig. 3:

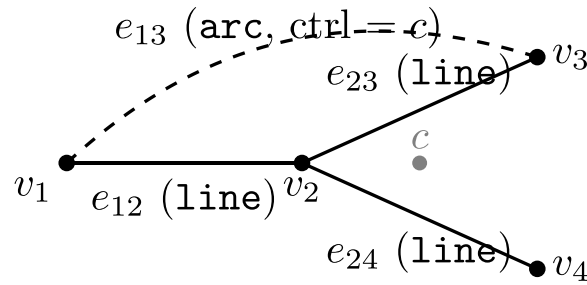


Fig. 4 Network primitives: vertices, edge kinds, and control vertices for curved edges

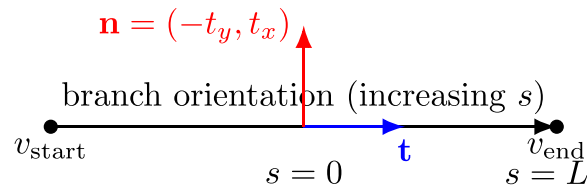


Fig. 5 Branch orientation and local tangent/normal convention

$$\pi^{(e)} = \begin{cases} +1, & \mathbf{e}_3^{(e)} \parallel +X_3 \quad (x_3 \text{ out of plane}), \\ -1, & \mathbf{e}_3^{(e)} \parallel -X_3 \quad (x_3 \text{ into plane}). \end{cases} \quad (1)$$

Vertices store coordinates and optional role labels (*node* or *control*), while edges connect two vertex IDs and may carry a polyline geometric parametrization (*line*, *arc*, *cubic_spline*) and optional control-vertex IDs for curved geometry. Topology is defined by endpoint connectivity only; control vertices affect geometry but not incidence. Input arrays support either implicit or explicit IDs:

$$\text{vertices} \in \mathbb{R}^{N_v \times 2} \text{ or } \mathbb{R}^{N_v \times 3}, \quad \text{connectivity} \in \mathbb{Z}^{N_e \times 2} \text{ or } \mathbb{Z}^{N_e \times 3} \quad (2)$$

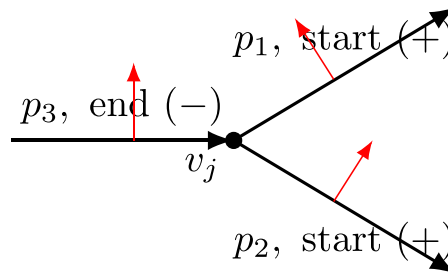
When IDs are omitted, consecutive IDs are assigned. Duplicate undirected edges are removed using the key $(\min(v_0, v_1), \max(v_0, v_1))$. Degree validation is enforced: $\deg(v) \leq N_{v, \max}$. Network primitives are schematically shown in Fig. 4.

Line edges are chained into polylines/branches. In dipolar crack mode, the connected components with the line-edge are ordered into paths (open chains or loops). In the polar crack mode, maximal branches are formed so that the interior vertices have a degree 2, and the branch endpoints are typically non-degree-2 vertices. Each branch is oriented from v_{start} to v_{end} , with arc length $s \in [0, L]$. Local frames follow

$$\mathbf{t}(s) = \frac{d\mathbf{x}}{ds}, \quad \mathbf{n}(s) = (-t_y, t_x) \quad (3)$$

i.e., $\mathbf{n}$ is the left-normal relative to increasing s . Conventions of branch orientation, tangent, and normal are depicted in Fig. 5.

Crack junctions are special topological features that require careful analysis. As shown in Fig. 6 junction jump unknowns are shared per junction vertex (*strict*), or defined per branch-end incidence (*core/soft*). The branch-face sign convention at a junction is $\text{sgn}(p, \text{start}) = +1$, $\text{sgn}(p, \text{end}) = -1$. Outgoing tangents are



$$\sum_{(p, \text{face}) \in \mathcal{I}(v_j)} \text{sgn}(p, \text{face}) (\cdot)_p = 0$$

Fig. 6 Junction incidence signs for branch faces: start +, end –

$$t_{\text{out}} = \begin{cases} t_{\text{mid}}(k=0), & \text{start incidence,} \\ -t_{\text{mid}}(k=N_p-1), & \text{end incidence.} \end{cases} \quad (4)$$

Network operations are implemented in the VCEM computational code as explicit graph transformations, geometric validity checks, and analysis/export utilities. Generation starts from the node positions sampled at $\mathbf{x}_i$ using one of three spatial modes: uniform, clustered, or grid. The junction nodes are assigned degree targets $d_i^* \in \{3, \dots, d_{\max}\}$, then connected by one of three strategies (nearest_neighbor, delaunay, random), followed by optional pruning to enforce $d_i \leq d_i^*$. Every candidate edge $e = (i, j)$ is checked against three geometric constraints before insertion: $\text{Intersect}(e, \tilde{e}) = \text{False} \quad \forall \tilde{e} \in E$. When intersections are disallowed, $d_{\min}(e, \tilde{e}) \geq \delta_{\min} \quad \forall \tilde{e} \in E$ for edge proximity control, and $\min(\theta, 180^\circ - \theta) \geq \theta_{\min}$ for junction-angle admissibility, where θ is the angle between the new edge direction and each existing incident edge at a junction.

Topological growth operations include: (i) adding tips to under-connected junctions, (ii) adding isolated tip–tip cracks, and (iii) creating connected polyline crack paths. Intersection processing is implemented in either a `single_pass` or `iterative` mode. For each true interior intersection between two edges, a new junction node v_k is inserted at $\mathbf{x}_k$, both original edges are removed, and each is split into two sub-edges; all affected lengths are updated from Euclidean distances. Boundary operations are provided by `Boundary` and `BoundaryManager`. Supported boundary primitives include linear segments, circular arcs, and imported CAD curves (Bezier/spline approximations). The manager can: add boundary edges/nodes to the crack graph, trim crack edges against a closed boundary, insert boundary-intersection nodes where needed, remove isolated non-boundary nodes, and filter candidate positions to interior points only. Import/export operations include boundary creation from DXF/SVG/text/STL/numpy arrays, and export to DXF/text. Network-level export returns arrays (vertices, connectivity, vertex_types) with sequential renumbering suitable for downstream solver interfaces. Finally, analysis operations include statistics and diagnostics: vertex/edge counts, type counts, degree distribution, edge-length moments, connected-component sizes, crack density, connectivity index, path lengths, and junction-angle sets.

Displacement and stress fields

Displacement discontinuity representation

We represent crack-face displacement discontinuities using distributions of edge dislocations aligned with local element axes, as illustrated in Fig. 7. The Burgers density in element e is

$$b^{(e)}(\xi) = b_1^{(e)}(\xi) e_1^{(e)} + b_2^{(e)}(\xi) e_2^{(e)}, \quad (5)$$

where Mode II corresponds to b_1 and Mode I to b_2 . The Burgers density is defined as the derivative of the displacement jump field along the crack arc. Specifically, we define

$$\frac{dJ}{ds}(s) = b_{II}(s) t(s) + b_I(s) n(s), \quad (6)$$

where $t(s)$ and $n(s)$ are the local unit tangent and normal vectors to the crack, respectively. The displacement jump vector is then obtained by integration,

$$J(s) = \int_0^s (b_{II}(\tilde{s}) t(\tilde{s}) + b_I(\tilde{s}) n(\tilde{s})) d\tilde{s}. \quad (7)$$

The crack opening displacement (COD) and crack sliding displacement (CSD) follow as projection: $COD(s) = J(s) \cdot n(s)$, $CSD(s) = J(s) \cdot t(s)$.

Several representations of the Burgers density $b_I(s), b_{II}(s)$ are used. In the Uniform Element Representation, we let $\{s_k\}_{k=0}^{N_p}$ denote the element nodes with $s_k = \frac{k}{N_p}L$, $k = 0, \dots, N_p$. Element midpoints are defined as

$$s_{k+\frac{1}{2}} = \frac{1}{2}(s_k + s_{k+1}). \quad (8)$$

The Burgers densities are taken piecewise-constant on each element:

$$b_I(s) = \sum_{k=0}^{N_p-1} \hat{b}_{I,k} \chi_k(s), \quad b_{II}(s) = \sum_{k=0}^{N_p-1} \hat{b}_{II,k} \chi_k(s), \quad (9)$$

where $\chi_k(s)$ is the characteristic function of the interval $[s_k, s_{k+1})$.

In the Chebyshev-Clustered Element Representation (`cheb_quad`), and to improve near-tip resolution, Chebyshev-Lobatto clustered nodes are used in normalized coordinates:

$$\xi_j = \frac{1}{2} \left(1 - \cos \frac{j\Phi}{N_p} \right), \quad j = 0, \dots, N_p. \quad (10)$$

The mapping to the physical segment $[s_a, s_b]$ gives the following.

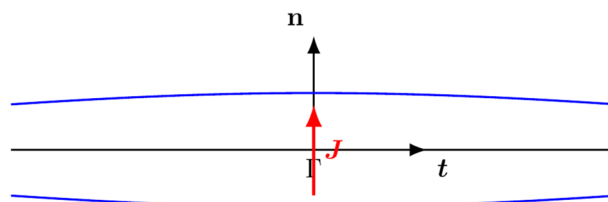


Fig. 7 Displacement jump $J = u^+ - u^-$ and its local components COD and CSD along a crack

$$s_j = s_a + (s_b - s_a) \xi_j. \quad (11)$$

The Burgers densities remain element wise constant as in (9), but collocation is performed at element midpoints.

Finally, in the Singular Representation (`singular`), we embed the correct square-root tip asymptotics, we factor the Burgers density as

$$b_I(s) = \hat{b}_I(s) \text{sing}(s), \quad b_{II}(s) = \hat{b}_{II}(s) \text{sing}(s), \quad (12)$$

where $\hat{b}_I, \hat{b}_{II}$ are regularized unknown densities.

For a full (dipolar) crack with two tips,

$$\text{sing}(s) = \frac{1}{\sqrt{s(L-s)}}. \quad (13)$$

For a half (polar) crack with a single tip at $s = L$,

$$\text{sing}(s) = \frac{1}{\sqrt{L-s}}. \quad (14)$$

Quadrature weights are scaled accordingly,

$$w_q^{\text{eff}} = w_q \text{sing}(s_q). \quad (15)$$

After solving for the regularized coefficients $\hat{b}_{I,k}, \hat{b}_{II,k}$, the physical Burgers densities at the midpoints of the elements are recovered as

$$b_{I,k} = \hat{b}_{I,k} \text{sing}(s_{k+\frac{1}{2}}), \quad b_{II,k} = \hat{b}_{II,k} \text{sing}(s_{k+\frac{1}{2}}). \quad (16)$$

This representation explicitly embeds the near-tip asymptotic behavior and is preferred for stress-intensity-factor extraction and crack propagation studies, at the cost of increased numerical sensitivity near endpoints. The crack kinematics are represented in terms of the displacement jump $[[u]] = u^+ - u^-$ across the crack faces. The Burgers-vector density therefore parameterizes the relative discontinuity, not the two face displacements separately, and no implicit symmetry assumption is required.

Crack polarity and burgers content vector

For a polyline element k with length Δs_k , tangent $\mathbf{t}_k$, and normal $\mathbf{n}_k$, the incremental jump is

$$\Delta \mathbf{J}_k = (b_{II}^k \mathbf{t}_k + b_I^k \mathbf{n}_k) \Delta s_k. \quad (17)$$

The accumulated jump at node m is

$$\mathbf{J}(s_m) = \sum_{k=0}^{m-1} \Delta \mathbf{J}_k, \quad (18)$$

which is a discrete approximation of (7). The COD and CSD follow.

Let Γ denote a crack segment parameterized by arc-length $s \in [0, L]$. We define the *Burgers content vector* of the crack as the line integral of the Burgers density along Γ ,

$$B = \int_{\Gamma} (b_{II}(s) t(s) + b_I(s) n(s)) ds. \quad (19)$$

Using (6), it follows directly that $B = J(L) - J(0)$, so that the Burgers content represents the net displacement discontinuity between the two crack endpoints. A crack is said to be *dipolar* (or Burgers-neutral) if $B = 0$. In this case, the total Burgers content vanishes, which is equivalent to

$$\int_{\Gamma} b_I(s) ds = 0, \quad \int_{\Gamma} b_{II}(s) ds = 0. \quad (20)$$

Physically, a dipolar crack does not carry a net Burgers vector and therefore does not induce a rigid-body displacement offset between its endpoints. Classical internal cracks in infinite or bounded elastic bodies satisfy this neutrality condition.

A crack is said to be *polar* if $B \neq 0$. In this case, the crack carries a nonzero net Burgers vector, producing a finite jump between its endpoints. Such cracks behave analogously to dislocation sources or termination defects and are typically associated with half-cracks, boundary-terminated cracks, or defects with prescribed displacement offsets. Thus, crack polarity is completely characterized by the Burgers content vector B : dipolar cracks are Burgers-neutral, while polar cracks carry a nonzero net Burgers vector.

A crack *kink* is a geometric turning point along a single crack path where the tangent direction is discontinuous but the crack remains a single connected curve. Topologically, the vertex has degree two in the crack graph. By contrast, a *junction* (e.g., γ -, T-, or star-node) is a point where three or more crack segments meet; the corresponding vertex has degree $m \geq 3$. In graph-theoretic terms, a kink preserves a single edge connectivity, whereas a junction introduces branching and thus alters the network topology. From the standpoint of elasticity, a kink represents a change of local basis along one crack, while a junction represents the interaction of multiple Burgers-density fields meeting at a common material point. The latter therefore introduces compatibility conditions analogous to conservation laws in networks.

Let $\Gamma^{(b)}$ denote the crack branch b with local tangent $e_1^{(b)}$, normal $e_2^{(b)}$ and polarity $\pi^{(e)} = \pm 1$. The local crack Burgers content vector is written as:

$$B^{(b)} = \pi^{(e)} (\Delta u_{II}^{(b)} e_1^{(b)} + \Delta u_I^{(b)} e_2^{(b)}) \quad (21)$$

(a) *Kinks*. At a kink, the displacement field must remain single-valued. Since the crack is continuous, the Burgers density representation is transported across the change in basis. The condition is therefore a continuity requirement: the limiting displacement field approaching the kink from either side must coincide. No additional conservation constraint arises because only two crack segments meet and share a common parametrization.

(b) *Junctions*. At a junction where $m \geq 3$ branches meet, displacement single-valuedness implies a closure condition. Let x_0 denote the junction point. Then the net Burgers vector must vanish, i.e., $\sum_{b=1}^m B^{(b)}(x_0) = 0$. In two-dimensional elasticity this yields exactly two independent scalar constraints (the x - and y -components). Physically,

this ensures that the junction does not act as a spurious dislocation source and that the elastic energy remains finite (Muskhelishvili 1953; Sih 1970). Without this closure condition, the displacement field becomes multi-valued and the problem ill-posed. It is worth mentioning that junction closure in crack networks is the discrete manifestation of a divergence-free topological flux, analogous to charge conservation in electromagnetism and Burgers-vector conservation in dislocation theory.

Elastic fields

Let $\mathbf{r} = \mathbf{x} - \boldsymbol{\xi}$ and $r = \|\mathbf{r}\|$, where $\mathbf{x} = (x_1, x_2)$ denotes the field point and $\boldsymbol{\xi} = (\xi_1, \xi_2)$ the source point. The Kelvin displacement tensor is

$$U_{ij}(\boldsymbol{\xi}, \mathbf{x}) = \frac{1}{8\pi\mu(1-\nu)} \left[-(\kappa + 1) \ln r \delta_{ij} + \frac{r_i r_j}{r^2} \right], \quad (22)$$

where μ is the shear modulus, ν is Poisson's ratio, δ_{ij} is the Kronecker delta, and $\kappa = 3 - 4\nu$ for plane strain (or $\kappa = \frac{3-\nu}{1+\nu}$ for plane stress). Equation (22) represents the fundamental solution of the two-dimensional Navier–Cauchy equations for linear isotropic elasticity, giving the i -th displacement component at $\mathbf{x}$ due to a unit point force applied in the j -direction at $\boldsymbol{\xi}$. The logarithmic singularity reflects the long-range nature of elastic interactions in two dimensions, while the anisotropic term $r_i r_j / r^2$ captures the directional coupling between source and field coordinates. This tensor serves as the Green's function underlying boundary integral and dislocation-based formulations and forms the kernel through which distributed Burgers densities generate displacement and stress fields in the present variational framework.

The traction at $\mathbf{x}$ on a boundary (or crack face) with outward unit normal $\mathbf{n}(\mathbf{x})$ is $t_i^{(j)}(\mathbf{x}, \boldsymbol{\xi}) = \sigma_{ik}^{(j)}(\mathbf{x}, \boldsymbol{\xi}) n_k(\mathbf{x})$. This defines the *traction kernel* (Kelvin traction fundamental solution).

$$T_{ij}(\mathbf{x}, \boldsymbol{\xi}) := t_i^{(j)}(\mathbf{x}, \boldsymbol{\xi}) = \sigma_{ik}^{(j)}(\mathbf{x}, \boldsymbol{\xi}) n_k(\mathbf{x}) = C_{ik\ell m} \frac{\partial U_{mj}(\mathbf{x}, \boldsymbol{\xi})}{\partial x_\ell} n_k(\mathbf{x}). \quad (23)$$

Let $\hat{\mathbf{r}} = \mathbf{r}/r$ and define the scalar $\eta = \hat{\mathbf{r}} \cdot \mathbf{n}(\mathbf{x})$. After straightforward contraction of indices in (23), one obtains the explicit expression

$$T_{ij}(\mathbf{x}, \boldsymbol{\xi}) = \frac{1}{8\pi(1-\nu)r} [-(\kappa - 1) \eta \delta_{ij} + 2 \hat{r}_i n_j + 2 \hat{r}_j n_i - 4 \eta \hat{r}_i \hat{r}_j], \quad (24)$$

where $n_i = n_i(\mathbf{x})$ and $\hat{r}_i = (x_i - \xi_i)/r$. Equation (24) is the Kelvin traction fundamental solution used in Somigliana-type boundary integral formulations; it is singular of order $1/r$ and must be interpreted in the Cauchy principal-value sense when $\mathbf{x} \rightarrow \boldsymbol{\xi}$ on Γ .

The displacement field in an elastic body occupying Ω with internal displacement discontinuities can be written, via Betti's reciprocal identity, as

$$u_i(\mathbf{x}) = u_i^\infty(\mathbf{x}) + \int_\Gamma [T_{ik}(\mathbf{x}, \boldsymbol{\xi}) \llbracket u_k \rrbracket(\boldsymbol{\xi}) - U_{ik}(\mathbf{x}, \boldsymbol{\xi}) t_k(\boldsymbol{\xi})] d\Gamma_\xi, \quad (25)$$

where U_{ik} is Kelvin's displacement tensor, T_{ik} is the corresponding traction kernel, $\llbracket u_k \rrbracket$ denotes the displacement jump across Γ , and t_k is the traction acting on the discontinuity. For a traction-free crack, $t_k = 0$, so only the first term remains. In the Volterra representation, the displacement jump is expressed in terms of a Burgers vector density $b_k(\boldsymbol{\xi})$,

defined as $\llbracket u_k \rrbracket(\boldsymbol{\xi}) = b_k(\boldsymbol{\xi})$. The traction kernel T_{ik} is obtained by differentiating Kelvin's solution in the direction of the outward normal $n_\ell(\boldsymbol{\xi})$, $T_{ik}(\mathbf{x}, \boldsymbol{\xi}) = \frac{\partial U_{ik}(\mathbf{x}-\boldsymbol{\xi})}{\partial \xi_\ell} n_\ell(\boldsymbol{\xi})$. Substituting this expression into (25) yields

$$u_i(\mathbf{x}) = u_i^\infty(\mathbf{x}) + \int_\Gamma \frac{\partial U_{ik}(\mathbf{x}-\boldsymbol{\xi})}{\partial \xi_\ell} n_\ell(\boldsymbol{\xi}) b_k(\boldsymbol{\xi}) d\Gamma_\xi, \quad (26)$$

Thus, the distributed Burgers density acts as a continuous distribution of Volterra dislocations, each contributing through spatial derivatives of Kelvin's fundamental solution. The stress field follows by applying the constitutive relation and spatial differentiation to the displacement representation.

$$\sigma_{ij}(\mathbf{x}) = \sigma_{ij}^\infty(\mathbf{x}) + \int_\Gamma C_{ijkl} \frac{\partial}{\partial x_l} \left(\frac{\partial U_{k\alpha}(\mathbf{x}-\boldsymbol{\xi})}{\partial \xi_m} n_m(\boldsymbol{\xi}) \right) b_\alpha(\boldsymbol{\xi}) d\Gamma_\xi. \quad (27)$$

Defining the elastostatic dislocation kernel

$$K_{ij\alpha}(\mathbf{x}, \boldsymbol{\xi}) = C_{ijkl} \frac{\partial}{\partial x_l} \left(\frac{\partial U_{k\alpha}(\mathbf{x}-\boldsymbol{\xi})}{\partial \xi_m} n_m(\boldsymbol{\xi}) \right),$$

we obtain the stress representation

$$\sigma_{ij}(\mathbf{x}) = \sigma_{ij}^\infty(\mathbf{x}) + \int_\Gamma K_{ij\alpha}(\mathbf{x}, \boldsymbol{\xi}) b_\alpha(\boldsymbol{\xi}) d\Gamma_\xi, \quad (28)$$

Hence, both the displacement and stress fields of a crack network are generated entirely by spatial derivatives of Kelvin's tensor, demonstrating that the dislocation-based formulation is fundamentally a boundary integral representation built upon the Green's function of the Navier–Cauchy operator.

The Burgers density components b_α are expressed in the global Cartesian frame. Define the relative coordinates $\Delta_i = x_i - \xi_i$, $R^2 = \Delta_1^2 + \Delta_2^2$, so that $\boldsymbol{\Delta} = \mathbf{x} - \boldsymbol{\xi}$. With this convention, define $M = \frac{2\mu}{\pi(\kappa+1)}$, which reduces to $M = \frac{\mu}{2\pi(1-\nu)}$, plane strain, $\frac{\mu(1+\nu)}{2\pi}$, plane stress, where μ is the shear modulus. The two-dimensional static stress kernel can then be written compactly as

$$K_{ij\alpha}(\mathbf{x}, \boldsymbol{\xi}) = M \left[\frac{\delta_{i\alpha} \Delta_j + \delta_{j\alpha} \Delta_i - \delta_{ij} \Delta_\alpha}{R^2} - \frac{2 \Delta_i \Delta_j \Delta_\alpha}{R^4} \right], \quad (29)$$

where δ_{ij} denotes the Kronecker delta. For an edge dislocation with the Burgers vector $\mathbf{b} = (b_1, b_2)$ aligned with the global axes, the stress field becomes

$$\sigma_{ij}(\mathbf{x}) = \sum_{\alpha=1}^2 K_{ij\alpha}(\mathbf{x}, \boldsymbol{\xi}) b_\alpha, \quad (30)$$

which yields the explicit expressions;

$$\begin{aligned}
\sigma_{11} &= -M \left[b_1 \frac{\Delta_2(3\Delta_1^2 + \Delta_2^2)}{R^4} - b_2 \frac{\Delta_1(\Delta_1^2 - \Delta_2^2)}{R^4} \right], \\
\sigma_{22} &= M \left[b_1 \frac{\Delta_2(\Delta_1^2 - \Delta_2^2)}{R^4} - b_2 \frac{\Delta_1(\Delta_1^2 + 3\Delta_2^2)}{R^4} \right], \\
\sigma_{12} &= M \left[b_1 \frac{\Delta_1(\Delta_1^2 - \Delta_2^2)}{R^4} + b_2 \frac{\Delta_2(\Delta_1^2 - \Delta_2^2)}{R^4} \right].
\end{aligned} \tag{31}$$

These expressions are the classical Volterra edge-dislocation stresses (Hirth and Lothe 1982). For Mode I loading, the Burgers vector is aligned with the crack normal, so that in a local frame $\mathbf{b} = b_2 \mathbf{e}_2$. Substituting $b_1 = 0$ into (31) yields

$$\sigma_{22}(x_1, x_2; \boldsymbol{\xi}) = -M b_2 \frac{\Delta_1(\Delta_1^2 + 3\Delta_2^2)}{R^4}. \tag{32}$$

If the crack lies along the x_1 axis and the source is located at $\xi_2 = 0$, then approaching the crack line from above ($x_2 \rightarrow 0^+$) gives the Cauchy-type singular kernel

$$K_{22}(x_1, 0^+; \xi_1) = -M \text{p.v.} \left(\frac{1}{x_1 - \xi_1} \right), \tag{33}$$

which is the kernel used in the crack-face traction equations, where “p.v.” denotes the Cauchy principal value.

For a collocation point $\mathbf{x}_i$, the discrete influence coefficient associated with the degree of freedom $\hat{b}_a^{(e)}$ takes the form

$$A_{i,a}^{(e)} = \int_{\Gamma_e} K(\mathbf{x}_i, \boldsymbol{\xi}) \pi^{(e)} N_a(\boldsymbol{\xi}) d\Gamma_{\boldsymbol{\xi}}, \tag{34}$$

where $K(\mathbf{x}, \boldsymbol{\xi})$ denotes the appropriate scalar kernel obtained by contracting the tensor kernel K_{ijk} (Eq. (29)) with the selected stress and Burgers components. For example, when computing $\sigma_{pq}(\mathbf{x}_i)$ due to Burgers component b_k , one has $K = K_{pqk}$. If a piecewise-constant approximation is employed, then $n_e = 1$ and $N_1(\boldsymbol{\xi}) \equiv 1$, so that (34) reduces to

$$A_{i,a}^{(e)} = \pi^{(e)} \int_{\Gamma_e} K(\mathbf{x}_i, \boldsymbol{\xi}) d\Gamma_{\boldsymbol{\xi}}.$$

Saddle point constrained variational formulation

Although [Crack network topology](#) and [Displacement and stress fields](#) section introduce several ingredients in a unified notation, many of these components are individually standard: graph-based crack/network descriptions are known in related settings, Burgers-density representations of crack opening follow classical distributed-dislocation formulations, and the elastic fields derive from established Kelvin fundamental solutions. The main novelty of the present work lies not in these ingredients taken separately, but in their integration into a constrained variational equilibrium framework for interacting crack networks, together with the associated KKT structure, junction-closure constraints, unilateral crack-face conditions, and finite-domain coupling, as will be presented next.

The energetic framework introduced in this section is used to formulate the constrained equilibrium problem for a prescribed crack geometry and to identify the

associated configurational driving force. It is not used here as a complete evolution principle over admissible crack sets. In the present work, crack growth is introduced separately through local near-tip propagation criteria after the equilibrium solution has been obtained for the current configuration.

Thermodynamic equilibrium

The variational formulation of fracture mechanics can be understood within the general framework of Legendre transforms, which underlie thermodynamic potentials and statistical ensembles. This viewpoint clarifies the distinction between force-controlled and displacement-controlled loading and provides a precise analogy between fracture evolution and phase transitions.

Let $f(x)$ be a differentiable function and define the conjugate variable $p = df/dx$. The Legendre transform of f with respect to x is: $g(p) = f(x) - px$, where x is now regarded as a function of p . The transformation replaces the variable x with its conjugate p . In thermodynamics, this procedure generates the various thermodynamic potentials from the internal energy. For a simple compressible system, $d\mathcal{E} = T dS - p dV$, the conjugate pairs are (T, S) and (p, V) . Performing Legendre transforms yields the Helmholtz ($\mathcal{A}$) and Gibbs (G) free energies: $\mathcal{A}(T, V) = \mathcal{E} - TS$, $G(T, p) = \mathcal{E} - TS + pV$.

In the present VCEM setting, the state of the system is described by the crack-field degrees of freedom $\mathbf{q}$ together with a geometric descriptor α of the crack network, so that the stored elastic energy is written as: $\mathcal{E} = \mathcal{E}(\mathbf{q}, \alpha)$. If the loading is prescribed in force form, the appropriate potential includes the external work,

$$\Phi(\mathbf{u}, \mathbf{q}, \alpha; \ell) = \mathcal{E}(\mathbf{u}, \mathbf{q}, \alpha) - \langle \ell, \mathbf{u} \rangle, \quad (35)$$

where ℓ denotes the applied dead load. The corresponding fracture functional is then

$$\mathcal{F}_{\text{force}}(\mathbf{u}, \mathbf{q}, \alpha) = \mathcal{E}(\mathbf{u}, \mathbf{q}, \alpha) - \langle \ell, \mathbf{u} \rangle + \Gamma A(\alpha). \quad (36)$$

Under displacement control, where the boundary displacement is prescribed and the external work is not an independent variable, the governing functional becomes

$$\mathcal{F}_{\text{disp}}(\mathbf{u}, \mathbf{q}, \alpha) = \mathcal{E}(\mathbf{u}, \mathbf{q}, \alpha) + \Gamma A(\alpha). \quad (37)$$

Thus, in either loading mode, equilibrium is determined by minimization over the full admissible state, not by the crack measure alone.

For a prescribed crack geometry, the reduced potential is defined by

$$\widehat{\Phi}(\alpha) = \min_{(\mathbf{u}, \mathbf{q}) \in \mathcal{A}(\alpha)} \Phi(\mathbf{u}, \mathbf{q}, \alpha). \quad (38)$$

The associated configurational driving force for crack change is obtained by differentiation with respect to the geometry,

$$\mathcal{G}[\alpha; \dot{\alpha}] = -D_{\alpha} \widehat{\Phi}(\alpha)[\dot{\alpha}], \quad (39)$$

where $D_{\alpha} \widehat{\Phi}(\alpha)[\dot{\alpha}]$ denotes the Gâteaux derivative in the admissible direction $\dot{\alpha}$. If the evolution is restricted to a one-parameter family $\alpha = \alpha(s)$, this reduces to

$$G(s) = -\frac{d\hat{\Phi}}{ds}, \quad G_A = -\frac{d\hat{\Phi}/ds}{dA/ds}. \quad (40)$$

The Griffith condition is then recovered in the usual form, $G_A = \Gamma$.

Because the Legendre structure differs under force and displacement control, the stability landscape differs as well. Under force control, increases in compliance lead to reductions in Φ , often producing snap-back or discontinuous energy drops, resembling a first-order phase transition. Under displacement control, energy varies more smoothly, allowing crack growth to occur in a more continuous fashion, analogous to a second-order or critical transition.

Thus, fracture mechanics admits a precise ensemble interpretation: different loading conditions correspond to different Legendre-transformed potentials, and crack growth corresponds to the evolution of an order parameter that minimizes the appropriate free energy functional.

The crack network may be characterized by an order parameter α such as:

$$\alpha = \frac{\sum_i \ell_i}{A} \quad (2D \text{ crack-length density}), \quad (41)$$

where $\sum_i \ell_i$ is the total network crack length, and A is the domain area. Alternatively, connectivity measures (e.g., spanning cluster fraction P_∞) can be used as an order parameter. For small α , cracks are isolated, and elastic interactions remain localized. As α increases, long-range interactions encoded in $\Phi(\alpha)$ promote coalescence. At a critical density α_c , a system-spanning crack may form, corresponding to a qualitative change in the structure of $\Phi(\alpha)$ and $\mathcal{F}(\alpha)$. Two limiting behaviors arise:

- **Continuous transition (percolation-like):** α increases gradually, $\Phi(\alpha)$ decreases smoothly, and connectivity grows continuously.
- **Discontinuous localization (first-order-like):** $\mathcal{F}(\alpha)$ develops multiple local minima, and a sudden jump to a dominant crack configuration occurs, accompanied by an abrupt energy release.

Fracture in interacting crack networks can therefore be interpreted as a hierarchical variational problem: an inner constrained elastic equilibrium by minimizing Φ together with its kinematic constraints, and an outer evolution driven by the free-energy landscape $\mathcal{F}(\alpha)$. This framework unifies classical Griffith fracture, network topology, and constrained optimization within a single thermodynamically consistent formulation. The order parameter (α) plays the role of a phase variable similar to phase density or magnetization, while the control or *tuning* parameters of external force or imposed displacement act as the roles of temperature and pressure in phase transition theory. The formulation presented here provides a description of constrained elastic equilibrium at the micro-scale, with reduced thermodynamic energy and an order parameter (α) governing evolution at the macro-scale.

Elastic energy functional and quadratic structure

The elastic energy stored in the crack system may be expressed as the work required to assemble the displacement discontinuity generated by the Burgers density against the induced traction field. Let $b_\alpha(\xi)$ denote the Burgers density components

resolved in the local crack basis $\{e_\alpha(\xi)\}$, so that the global Burgers vector reads $b_j(\xi) = e_{\alpha j}(\xi) b_\alpha(\xi)$, $b(\xi) = b_\alpha(\xi) e_\alpha(\xi)$. The traction field induced on the crack by a distributed Burgers density is written in global form as

$$t_i(x) = \int_{\Gamma} \mathcal{T}_{i\alpha}(x, \xi) b_\alpha(\xi) d\Gamma_\xi, \quad (42)$$

where $\mathcal{T}_{i\alpha}$ is the traction kernel derived from the dislocation kernel $K_{ij\alpha}$ via $\mathcal{T}_{i\alpha}(x, \xi) = K_{ij\alpha}(x, \xi) n_j(x)$, with $n_j(x)$ the outward unit normal at the field point. The index α is not to be confused with the *order parameter* defined earlier. The local traction components are obtained by projection $t_\beta(x) = e_{\beta i}(x) t_i(x)$. Substituting (42) into this projection gives

$$t_\beta(x) = \int_{\Gamma} \underbrace{e_{\beta i}(x) \mathcal{T}_{i\alpha}(x, \xi)}_{\mathcal{T}_{\beta\alpha}(x, \xi)} b_\alpha(\xi) d\Gamma_\xi, \quad (43)$$

where the local-to-local interaction operator is defined by

$$\mathcal{T}_{\beta\alpha}(x, \xi) := e_{\beta i}(x) \mathcal{T}_{i\alpha}(x, \xi). \quad (44)$$

This operator maps the Burgers component α at ξ to the traction component β at x .

Let the crack network geometry be denoted by $\Gamma(\alpha)$, where α represents a geometric or topological order parameter (e.g., crack density, total crack length per unit area, or another network descriptor). For a fixed configuration $\Gamma(\alpha)$, the internal elastic state is described by generalized degrees of freedom $q \in \mathbb{R}^n$ (e.g., Burgers density coefficients).

The total potential energy of the elastic body is

$$\Phi(q; \alpha) = \mathcal{E}(q; \alpha) - \mathcal{W}(q; \alpha), \quad (45)$$

where $\mathcal{E}$ is the stored elastic energy and $\mathcal{W}$ is the external work. For linear elasticity with distributed dislocation representation, $\Phi(q; \alpha)$ is quadratic in q . The elastic energy functional is then given by

$$\Phi[b] = \frac{1}{2} \int_{\Gamma} b_\beta(x) t_\beta(x) d\Gamma_x - \int_{\Gamma} b_\beta(x) t_\beta^{\text{app}}(x) d\Gamma_x, \quad (46)$$

where t_β^{app} denotes the applied traction resolved in the same local basis.

Substituting (43) into (46) yields the quadratic form

$$\Phi[b] = \frac{1}{2} \int_{\Gamma} \int_{\Gamma} b_\beta(x) \mathcal{T}_{\beta\alpha}(x, \xi) b_\alpha(\xi) d\Gamma_\xi d\Gamma_x - \int_{\Gamma} b_\beta(x) t_\beta^{\text{app}}(x) d\Gamma_x. \quad (47)$$

Under the standard assumptions of linear isotropic elasticity, the bilinear form defined by $\mathcal{T}_{\beta\alpha}(x, \xi)$ is symmetric in the sense of Betti reciprocity, ensuring that $\Phi[b]$ is a convex quadratic functional in the absence of inequality constraints.

Two equivalent viewpoints are used in boundary-integral crack mechanics. (i) The *strong-form* statement prescribes that the on-crack traction induced by the Burgers density balances the applied traction, yielding the Somigliana traction integral equation in principal-value form. (ii) The *variational* statement minimizes the elastic potential energy expressed in terms of the Burgers density. In the absence of inequality constraints, these are equivalent: the Somigliana equation is the Euler–Lagrange condition

obtained by taking the first variation of the energy functional with respect to b . We adopt the variational form because it admits a natural incorporation of kinematic constraints, as will be discussed next. Taking the first variation of (47) gives

$$\delta\Phi[b; \delta b] = \int_{\Gamma} \delta b_{\alpha}(x) (t_{\alpha}[b](x) - t_{\alpha}^{\text{app}}(x)) \, d\Gamma_x.$$

Since δb is arbitrary (within the admissible space), stationarity $\delta\Phi = 0$ implies the strong-form traction balance

$$t_{\alpha}[b](x) = t_{\alpha}^{\text{app}}(x), \quad x \in \Gamma, \quad (48)$$

which is the Somigliana traction integral equation. When admissibility constraints are present (junction closure, gauge, and unilateral contact), the stationarity condition is replaced by optimization conditions in which constraint reactions modify the traction balance through additional multiplier terms. The variational problem solved in this work is defined for a fixed crack geometry. Topological changes such as extension or branching are not treated as differentiable variations of the same functional, but as separate geometric updates followed by re-resolution of the equilibrium problem.

Kinematic constraints and the saddle point problem

Because cracks introduce a number of topological constraints, simple energy minimization is not a suitable approach, and one has to use a constrained optimization framework. A robust formulation is that by Karush, Kuhn, and Tucker (KKT) (Karush 1939; Kuhn and Tucker 1951), as it forms the cornerstone of modern applications of optimization methods (Nocedal and Wright 2006; Boyd and Vandenberghe 2004). In the context of boundary integral methods for crack mechanics, constrained KKT-type systems arise naturally in the distributed dislocation technique, where each crack branch must satisfy integral compatibility conditions at junctions and free ends. Blandford, Ingraffea, and Liggett (1981) established two-dimensional boundary element formulations that implicitly impose such constraints. The distributed dislocation framework was systematized by Hills and Nowell (1994), whose formulation is fully compatible with the KKT structure developed here. The energy minimization point of view has also been used in a variational model of the evolution of quasistatic cracks (Francfort and Marigo 1998; Bourdin et al. 2000).

Physical crack networks must satisfy integral constraints that ensure:

1. **Crack closure:** At crack tips and free ends, the total opening displacement must vanish: $\int_{\Gamma_{\text{tip}}} b(s) \, ds = 0$.
2. **Junction compatibility:** At crack junctions where multiple branches meet, the opening displacements must be continuous. For a junction connecting branches $\Gamma_1, \Gamma_2, \dots, \Gamma_k$: $\sum_{i=1}^k \pi_i^{(b)} \int_{\Gamma_i} b(s) \, ds = 0$, where $\pi_i^{(b)}$ account for branch polarity.
3. **Gauge conditions:** To remove rigid-body modes, we may impose $\int_{\Gamma} b(s) \, ds = u^{\text{prescribed}}$, for specified displacement constraints.
4. **Inequality constraints** $g(q) \geq 0$ enforce the non-penetration of crack faces, expressed as the non-negativity of crack-opening displacement (COD): $\text{COD}(s) = n \cdot J(s) \geq 0$, where $J(s)$ is the displacement-jump field.

These constraints can be written compactly as a linear operator C acting on b :

$$C[b] = 0. \quad (49)$$

The constrained minimization problem is:

$$\min_{b \in \mathcal{B}} \Phi[b] \quad \text{subject to} \quad C[b] = 0, \quad (50)$$

where $\mathcal{B}$ is the space of admissible Burgers density fields.

We introduce a Lagrange multiplier field $\lambda(s)$ (or a discrete vector $\boldsymbol{\lambda} \in \mathbb{R}^m$ for constraints m) to enforce the constraints. The Lagrangian functional is:

$$\mathcal{L}[b, \boldsymbol{\lambda}] = \Phi[b] + \langle \boldsymbol{\lambda}, C[b] \rangle, \quad (51)$$

where $\langle \cdot, \cdot \rangle$ denotes an appropriate inner product (e.g., L^2 for continuous fields or the Euclidean dot product for discrete vectors).

Expanded, this becomes:

$$\mathcal{L}[b, \boldsymbol{\lambda}] = \frac{1}{2} \int_{\Gamma} \int_{\Gamma} b(s) \cdot \mathcal{T}(s, s') b(s') ds ds' - \int_{\Gamma} b(s) \cdot t^{\text{app}}(s) ds + \langle \boldsymbol{\lambda}, C[b] \rangle. \quad (52)$$

The first-order optimality conditions (Karush-Kuhn-Tucker conditions for the equality-constrained case) require stationarity with respect to both b and $\boldsymbol{\lambda}$:

$$\frac{\delta \mathcal{L}}{\delta b} = 0 \quad \Rightarrow \quad \int_{\Gamma} \mathcal{T}(s, s') b(s') ds' - t^{\text{app}}(s) + C^T[\boldsymbol{\lambda}](s) = 0, \quad (53)$$

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{\lambda}} = 0 \quad \Rightarrow \quad C[b] = 0, \quad (54)$$

where C^T is the adjoint operator of C . Equation (53) is the *Somigliana traction integral equation* modified by the constraint reaction forces $C^T[\boldsymbol{\lambda}]$, and equation (54) recovers the primal constraint. We discretize the Burgers density field using a finite element expansion:

$$b(s) \approx b_h(s) = \sum_{i=1}^n q_i N_i(s), \quad (55)$$

where:

- $\{q_i\}_{i=1}^n$ are the nodal degrees of freedom (DOFs),
- $\{N_i(s)\}_{i=1}^n$ are shape functions (basis functions),
- n is the total number of DOFs in the crack network.

Similarly, the constraint operator becomes a finite-dimensional matrix:

$$C[b_h] \approx Cq = 0, \quad C \in \mathbb{R}^{m \times n}, \quad q = [q_1, \dots, q_n]^T. \quad (56)$$

The Lagrange multipliers are discretized as:

$$\boldsymbol{\lambda} \approx \boldsymbol{\lambda}_h = [\boldsymbol{\lambda}_1, \dots, \boldsymbol{\lambda}_m]^T \in \mathbb{R}^m. \quad (57)$$

Substituting (55) into (52), the discrete Lagrangian becomes:

$$\mathcal{L}(q, \boldsymbol{\lambda}) = \Phi(q) + \boldsymbol{\lambda}^T Cq, \quad (58)$$

where $\Phi(q)$ is the discrete approximation of $\Phi[b]$. The specific form of $\Phi(q)$ depends on the discretization method: If the element interpolation is linear ($n_s = 2$), the standard Lagrange shape functions on $[-1, 1]$ are used. The Burgers density then varies linearly along the element (e),

$$b^{(e)}(\xi) = \frac{1-\xi}{2} q_1^{(e)} + \frac{1+\xi}{2} q_2^{(e)}. \quad (59)$$

This provides first-order accuracy and constant strain within the element. If a quadratic interpolation is used ($n_s = 3$), the Burgers density becomes

$$b^{(e)}(\xi) = \frac{1}{2}\xi(\xi-1)q_1^{(e)} + (1-\xi^2)q_2^{(e)} + \frac{1}{2}\xi(\xi+1)q_3^{(e)}. \quad (60)$$

Quadratic elements allow curvature in the Burgers density distribution and improve accuracy for crack-tip and junction stress fields without mesh refinement. Higher-order interpolations follow directly by increasing n_s and selecting appropriate polynomial or spectral bases (e.g., higher-order Lagrange or Chebyshev functions). The order of the shape functions controls the approximation quality of the crack-field representation and the convergence rate of the numerical scheme.

- **Galerkin (energy-based):**

$$\Phi_{\text{Gal}}(q) = \frac{1}{2}q^T A q - f^T q, \quad (61)$$

where A is the Galerkin stiffness matrix and f is the load vector.

- **Least-squares (residual-based):**

$$\Phi_{\text{LS}}(q) = \frac{1}{2}\|Kq - f\|^2 + \frac{\rho}{2}\|q\|^2, \quad (62)$$

where K is the collocation influence matrix, and $\rho \geq 0$ is a Tikhonov regularization parameter.

$$K_{ij} = \int_{\Gamma} \mathcal{T}(s, s') N_j(s') ds'. \quad (63)$$

Taking the gradient of $\mathcal{L}(q, \lambda)$ with respect to q and λ and setting it to zero yields the KKT conditions:

$$\nabla_q \mathcal{L} = \nabla_q \Phi(q) + C^T \lambda = 0, \quad (64)$$

$$\nabla_{\lambda} \mathcal{L} = Cq = 0. \quad (65)$$

The second equation enforces admissibility ($Cq = 0$), while the first expresses equilibrium between energetic forces $\nabla_q \Phi$ and the constraint reactions $C^T \lambda$. The pair (q^*, λ^*) therefore satisfies the stationarity of the Lagrangian. This system has a saddle-point structure. Indeed, the physical problem is $\min_{q \in \ker C} \Phi(q)$, i.e., minimization of energy over the admissible set. Introducing Lagrange multipliers converts the constrained minimization into the equivalent min-sup problem

$$\min_q \sup_{\lambda} \mathcal{L}(q, \lambda).$$

The supremum over λ enforces feasibility: if $Cq \neq 0$, then $\mathcal{L}(q, \lambda)$ is unbounded above in the direction of λ , so infeasible configurations are assigned infinite cost and are excluded from the outer minimization. Consequently, $\mathcal{L}$ is minimized with respect to the primal variables q (energy principle) and effectively maximized with respect to the dual variables λ (constraint enforcement).

At the stationary point, the Hessian of $\mathcal{L}$ has the block form

$$\begin{bmatrix} \nabla_q^2 \Phi & C^T \\ C & 0 \end{bmatrix},$$

which is symmetric but indefinite: curvature is positive in admissible q directions (energy convexity) and zero in pure λ directions, with coupling through C . This indefiniteness is characteristic of mixed and constrained variational formulations, and is the algebraic manifestation of exact constraint enforcement without penalty regularization.

Rearranging (64)–(65) into matrix form:

$$\begin{bmatrix} H & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} q \\ \lambda \end{bmatrix} = \begin{bmatrix} r \\ 0 \end{bmatrix}, \quad (66)$$

where:

- $H = \nabla_q^2 \Phi(q)$ is the Hessian of the objective function,
- r denotes the negative objective gradient evaluated at the linearization point (for a quadratic objective, $r = f$),
- The system size is $(n + m) \times (n + m)$.

Realization in Galerkin and least-squares

Galerkin formulation

For the Galerkin objective (61), we have:

$$\nabla_q \Phi_{\text{Gal}}(q) = Aq - f, \quad (67)$$

$$\nabla_q^2 \Phi_{\text{Gal}}(q) = A. \quad (68)$$

The KKT system becomes

$$\begin{bmatrix} A & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} q \\ \lambda \end{bmatrix} = \begin{bmatrix} f \\ 0 \end{bmatrix}, \quad (69)$$

where:

- $A \in \mathbb{R}^{n \times n}$ is the **Galerkin stiffness matrix**:

$$A_{ij} = \int_{\Gamma} N_i(s) \int_{\Gamma} \mathcal{T}(s, s') N_j(s') ds' ds. \quad (70)$$

This is symmetric ($A = A^T$) and positive semi-definite due to the symmetry of $\mathcal{T}$.

- $f \in \mathbb{R}^n$ is the **Galerkin load vector**:

$$f_i = \int_{\Gamma} N_i(s) \cdot t^{\text{app}}(s) ds. \quad (71)$$

- The total energy is $\Phi_{\text{Gal}}(\mathbf{q}) = \frac{1}{2} \mathbf{q}^T \mathbf{A} \mathbf{q} - \mathbf{b}^T \mathbf{q}$, which is the direct discrete analog of the continuous energy version.

The Galerkin formulation is energy-consistent: $\mathbf{q}^T \mathbf{A} \mathbf{q} = 2\Phi_{\text{elastic}}[\mathbf{b}_h]$, and its symmetric structure enables efficient Cholesky factorization. However, it requires double quadrature (over source and field elements) for assembly. The structure does not need regularization since constraints in $\mathbf{C}$ remove null space.

Least-squares formulation

For the least-squares objective (62), expanding the norm:

$$\Phi_{\text{LS}}(\mathbf{q}) = \frac{1}{2} (\mathbf{K} \mathbf{q} - \mathbf{f})^T (\mathbf{K} \mathbf{q} - \mathbf{f}) + \frac{\rho}{2} \mathbf{q}^T \mathbf{q} \quad (72)$$

$$= \frac{1}{2} \mathbf{q}^T (\mathbf{K}^T \mathbf{K} + \rho \mathbf{I}) \mathbf{q} - \mathbf{f}^T \mathbf{K} \mathbf{q} + \frac{1}{2} \mathbf{f}^T \mathbf{f}. \quad (73)$$

Taking derivatives:

$$\nabla_{\mathbf{q}} \Phi_{\text{LS}}(\mathbf{q}) = (\mathbf{K}^T \mathbf{K} + \rho \mathbf{I}) \mathbf{q} - \mathbf{K}^T \mathbf{f} = \mathbf{H} \mathbf{q} - \mathbf{K}^T \mathbf{f}, \quad (74)$$

$$\nabla_{\mathbf{q}}^2 \Phi_{\text{LS}}(\mathbf{q}) = \mathbf{K}^T \mathbf{K} + \rho \mathbf{I} = \mathbf{H}. \quad (75)$$

The KKT system becomes

$$\begin{bmatrix} \mathbf{H} & \mathbf{C}^T \\ \mathbf{C} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} \mathbf{K}^T \mathbf{f} \\ \mathbf{0} \end{bmatrix}, \quad \mathbf{H} = \mathbf{K}^T \mathbf{K} + \rho \mathbf{I}, \quad (76)$$

where:

- $\mathbf{f} \in \mathbb{R}^{N_c}$ is the applied traction vector.
- $\mathbf{H} = \mathbf{K}^T \mathbf{K} + \rho \mathbf{I}$ is the regularized normal equations Hessian, which is symmetric positive definite (SPD) by construction.
- The least-squares residual is $\mathbf{r}(\mathbf{q}) = \mathbf{K} \mathbf{q} - \mathbf{f}$, which measures traction mismatch at collocation points.

strong-form collocation enforces the traction equation pointwise, requires a single quadrature (over source elements only) and its regularization $\rho > 0$ ensures well-posedness for ill-conditioned $\mathbf{K}$. However, it is not energy-consistent in the same sense as Galerkin. Its Hessian $\mathbf{H}$ is SPD, but the full system (76) is indefinite.

Both formulations solve the same physical problem — finding the Burgers density distribution $\mathbf{b}(s)$ that satisfies the Somigliana integral equation subject to admissibility constraints, but they differ in how the continuous energy functional is discretized. The KKT framework unifies both: they share the constraint structure $\mathbf{C} \mathbf{q} = \mathbf{0}$ and multipliers $\boldsymbol{\lambda}$, differing only in the block $\mathbf{H}$ (or $\mathbf{A}$) and right-hand side. This allows for seamless switching between formulations in implementation. A comparison of Galerkin and least-squares KKT realizations is given in Table 1.

Table 1 Comparison of Galerkin and least-squares KKT realizations

Property	Galerkin	Least-Squares
Objective	Energy $\frac{1}{2}\mathbf{q}^T \mathbf{A} \mathbf{q} - \mathbf{b}^T \mathbf{q}$	Residual $\frac{1}{2}\ \mathbf{K} \mathbf{q} - \mathbf{f}\ ^2$
Hessian	$\mathbf{A}$ (symmetric PSD)	$\mathbf{H} = \mathbf{K}^T \mathbf{K} + \rho \mathbf{I}$ (symmetric SPD)
Assembly	Double quadrature	Single quadrature
Energy-consistent	Yes	No
Regularization	Not needed	Required ($\rho > 0$)
Sparsity	Dense	Dense
Solver	Cholesky after constraint reduction	Direct (LAPACK) or SVD

Junction and crack opening constraints

High degree junctions ($\deg \geq 3$) result in strong coupling between crack branches participating in the junction. This comes in two ways. First, through the stress interaction kernel of nearby nodes on branches (Eq. 29), and second, stronger coupling arises as a result of the strict enforcement of compatibility constraints. At a crack junction where $m \geq 3$ branches meet, single-valued displacement implies the Burgers closure condition $\sum_{b=1}^m \mathbf{B}(\mathbf{x}_0) = 0$. In discrete form, this becomes the linear compatibility constraint

$\mathbf{C}_J \mathbf{q} = 0$, where $\mathbf{C}_J \in \mathbb{R}^{p \times n}$ evaluates the branch crack openings at the junction node and assembles their global components. In two-dimensional elasticity, each junction contributes two scalar constraints. The crack DOF space, therefore, decomposes as $\mathbb{R}^n = \mathcal{N}(\mathbf{C}_J) \oplus \mathcal{R}(\mathbf{C}_J^T)$, where $\mathcal{N}(\mathbf{C}_J)$ is the compatible subspace, and $\mathcal{R}(\mathbf{C}_J^T)$ represents constraint reaction directions.

In this *strict* junction mode, the crack topology is strictly preserve and the conservation of the Burgers vector are maintained at every junction is guaranteed. The resulting system is symmetric but indefinite. In some simulations of crack network evolution, this strict condition can be relaxed to expedite calculations of network evolution. We then introduce two approximate modes, as below.

In a *core* junction mode, a small region around the junction is allowed to have a gap (void). This type of deformation is physically admissible and lies in the compatible subspace, $\mathbf{q} \in \mathcal{N}(\mathbf{C}_J)$, $\mathbf{C}_J \mathbf{q} = 0$. Only these modes contribute to elastic energy storage and govern stress intensity factors and crack growth. They exactly satisfy Burgers-vector conservation on each incident branch.

The *soft* junction mode is a compromise, where compatibility is enforced weakly by adding a quadratic penalty to the energy:

$$\Phi_\eta(\mathbf{q}) = \frac{1}{2} \mathbf{q}^T \mathbf{H} \mathbf{q} - \mathbf{f}^T \mathbf{q} + \frac{\eta}{2} \|\mathbf{C}_J \mathbf{q}\|^2,$$

where $\eta > 0$ is a penalty parameter. The corresponding normal equations become the following.

$$(\mathbf{H} + \eta \mathbf{C}_J^T \mathbf{C}_J) \mathbf{q} = \mathbf{f}.$$

Equivalently, in least-squares form,

$$\tilde{\mathbf{K}} = \begin{bmatrix} \mathbf{K} \\ \sqrt{\eta} \mathbf{C}_J \end{bmatrix}, \quad \tilde{\mathbf{f}} = \begin{bmatrix} \mathbf{f} \\ 0 \end{bmatrix},$$

leading to the Hessian

$$\tilde{\mathbf{H}} = \mathbf{K}^T \mathbf{K} + \eta \mathbf{C}_J^T \mathbf{C}_J + \rho \mathbf{I}.$$

Here, ρ denotes the optional Tikhonov regularization. In the soft mode, $\mathbf{C}_J \mathbf{q} \approx 0$, with residual magnitude controlled by η . The system is symmetric positive definite, but compatibility is only approximate. No additional constraint beyond displacement single-valuedness is required; the conservation law follows from kinematics. Strict enforcement satisfies this law identically, while soft enforcement satisfies it only approximately. Core modes are precisely those satisfying the conservation law.

The three junction modes are summarized in Table 2. The distinction is not merely algebraic; it reflects whether junction closure is treated as a conservation law or as a numerical stabilization device. In addition to junction closure, crack faces must satisfy the unilateral contact (non-penetration) condition, $\Delta u_I(\xi) \geq 0$, with equality at the contact points. This constitutes an inequality constraint in the variational problem. Let $\mathbf{C}_N \mathbf{q}$ denote the discretized normal crack opening evaluated in the candidate contact DOFs. The complementarity condition reads

$$\mathbf{C}_N \mathbf{q} \geq 0, \quad \boldsymbol{\mu} \geq 0, \quad \boldsymbol{\mu}^T (\mathbf{C}_N \mathbf{q}) = 0,$$

where $\boldsymbol{\mu}$ are the contact reaction multipliers. The full constrained problem, therefore, becomes.

$$\min_{\mathbf{q}} \frac{1}{2} \mathbf{q}^T \mathbf{H} \mathbf{q} - \mathbf{f}^T \mathbf{q} \quad \text{subject to} \quad \mathbf{C}_J \mathbf{q} = 0, \quad \mathbf{C}_N \mathbf{q} \geq 0.$$

Inequality constraints are handled by an active-set strategy. Let $\mathcal{A}$ denote the set of DOFs where $\Delta u_I = 0$. At a given iteration, the problem reduces to equality constraints:

$$\mathbf{C}_J \mathbf{q} = 0, \quad \mathbf{C}_N^{(\mathcal{A})} \mathbf{q} = 0.$$

The augmented KKT system becomes

$$\begin{bmatrix} \mathbf{H} & \mathbf{C}_J^T & \mathbf{C}_N^{(\mathcal{A})T} \\ \mathbf{C}_J & 0 & 0 \\ \mathbf{C}_N^{(\mathcal{A})} & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \boldsymbol{\lambda} \\ \boldsymbol{\mu} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ 0 \\ 0 \end{bmatrix}.$$

The active set is updated iteratively until the complementarity conditions are satisfied.

To clarify junction conditions, an example is shown in Fig. 8, we show the results of the VCEM code for the following conditions: $E=200$ GPa, $\nu = 0.30$, applied stress $\sigma_{yy} = 100$ MPa, number of elements=40, meshing=uniform, number of collocation points=3 and number of Gauss points=6. In the strict formulation, the junction constraint is enforced algebraically as $\mathbf{C}_J \mathbf{q} = 0$ through the KKT system. Let the compatibility residual be defined as $r_J := \|\mathbf{C}_J \mathbf{q}\|_2$. For an exact solution of the saddle-point

Table 2 Comparison of junction treatment modes

Mode	Mathematical form	Physical effect	Numerical properties
Core	$\mathbf{q} \in \mathcal{N}(\mathbf{C}_J)$	Physically admissible crack opening patterns; exact Burgers closure	Reduced compatible subspace; no constraint reactions
Strict	$\mathbf{C}_J \mathbf{q} = 0$ enforced via KKT	Exact conservation of Burgers vector; preserves topology and branching physics	Saddle-point system; symmetric but indefinite
Soft	Penalty $\frac{\eta}{2} \ \mathbf{C}_J \mathbf{q}\ ^2$	Approximate closure; small compatibility leakage if η finite	SPD system; conditioning controlled by η ; may introduce artificial stiffness

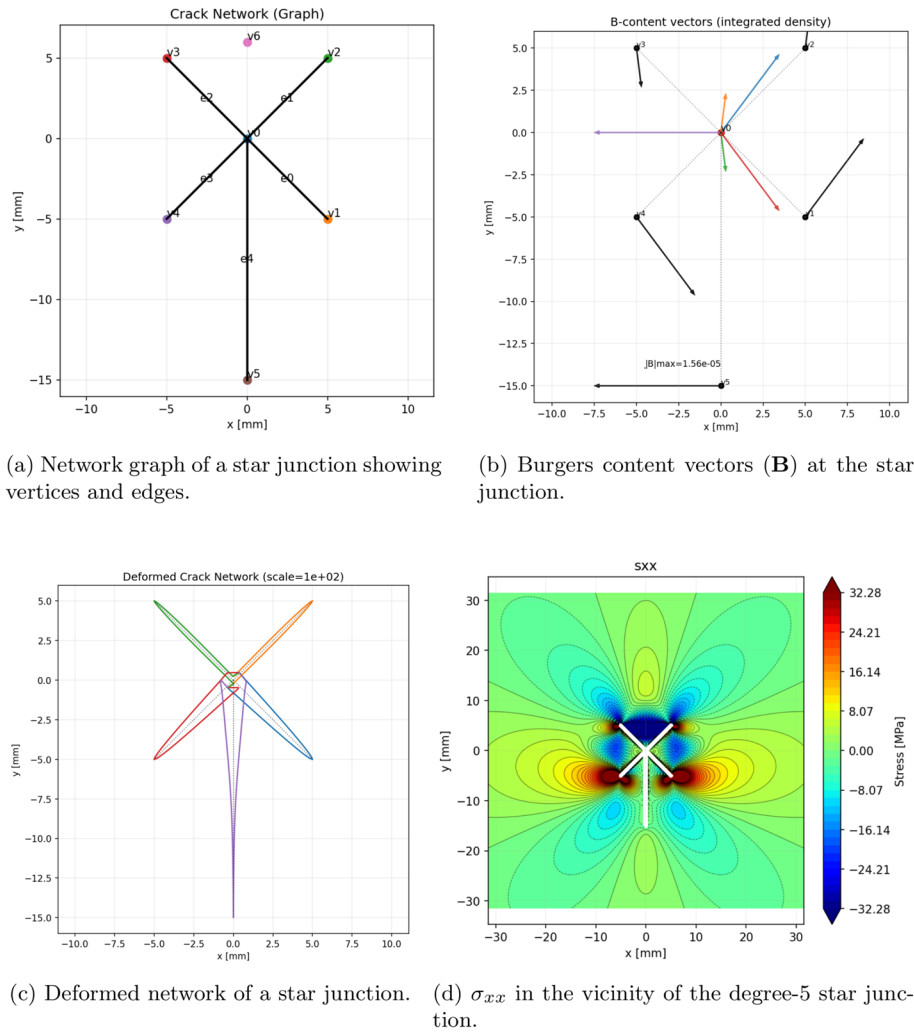


Fig. 8 Example of a star junction illustrating the *strict* junction mode. All branches share identical displacement (not COD) at the junction point. The algebraic sum of Burgers content vectors of the five branches vanishes at the junction, satisfying the closure condition

system, $r_J = 0$ in exact arithmetic. In floating-point computations, the residual satisfies $r_J = \mathcal{O}(\varepsilon_{\text{mach}})\|\mathbf{q}\|_2$, where $\varepsilon_{\text{mach}}$ denotes machine precision. For the star-junction example of Fig. 8, the numerical solution yields

$$r_J \leq \epsilon_{\text{lin}}\|\mathbf{f}\|_2,$$

where ϵ_{lin} is the linear solver tolerance. The computed Burgers closure, $\sum_{b=1}^5 \mathbf{B}^{(b)}(\mathbf{x}_0)$

' is therefore zero to within solver tolerance, consistent with the graphical cancellation observed in panel (b). Equivalently, the projection of $\mathbf{q}$ onto the constraint reaction space $\mathcal{R}(\mathbf{C}_J^T)$ is negligible, i.e., $\|\mathbf{P}_{\mathcal{R}}\mathbf{q}\|_2 \ll \|\mathbf{q}\|_2$, where $\mathbf{P}_{\mathcal{R}}$ denotes the orthogonal projector onto $\mathcal{R}(\mathbf{C}_J^T)$. This confirms that the computed solution lies entirely within the compatible subspace $\mathcal{N}(\mathbf{C}_J)$ up to machine precision. Consequently: (i) Burgers-vector conservation holds exactly, (ii) no artificial dislocation source is introduced at the node, and (iii) the stress field in Fig. 8d remains free of non-physical singularities associated

with compatibility violation. The quantitative smallness of r_J therefore provides algebraic confirmation that the strict junction mode preserves topological conservation and that the deformation shown in Fig. 8c is generated solely by admissible core modes.

The stress field in Fig. 8d clearly distinguishes between two topologically different vertices: degree-1 tips and the degree-5 junction. Each terminal branch (degree-1 vertices) exhibits the classical LEFM square-root singularity. The singular lobes are visible in Fig. 8d as the localized stress concentrations near each free crack end. The singularity arises from traction-free boundary conditions and is an intrinsic feature of crack tips.

In a kink (change of crack direction without branching), no additional singularity is introduced beyond the standard tip behavior $r^{-1/2}$. The displacement field remains single-valued, and the Burgers density is transported continuously across the kink. Consequently, the local asymptotic structure is governed by the usual mixed-mode expansion, with modified angular functions due to the change in crack orientation. No additional incompatibility singularity is observed. The stress field components of a kinked crack are shown in Fig. 9. By contrast, the central star junction (degree-5 vertex) does *not* exhibit a crack-tip singularity because the strict formulation enforces the Burgers closure condition. Hence, the junction does not act as a source of net dislocation content. The displacement field remains continuous and bounded, and the stress field near the junction behaves as $\sigma_{ij}(r) = \mathcal{O}(1)$ as $r \rightarrow 0$, rather than exhibiting an $r^{-1/2}$ singularity. This bounded behavior is evident in Fig. 8d: while stress redistributes due to branch interaction, no localized singular core appears at the central node. The absence of a junction singularity confirms that topological compatibility eliminates artificial stress sources. In effect, the strict junction behaves as an internal equilibrium node rather than as a crack tip.

Figure 9 shows the stress field around a kinked crack. Strong singular behavior is associated with crack tips, but not at the kink. Thus, singular stresses arise only at degree-1 vertices (true crack tips), are geometrically reoriented but not intensified at degree-2 kinks, and remain bounded at higher-degree junctions when Burgers closure is enforced strictly. If compatibility were relaxed (soft mode), a residual $C_{Jq} \neq 0$ would introduce a concentrated incompatibility, producing an artificial singular stress core at the junction. The strict formulation, therefore, guarantees that all physically admissible singular

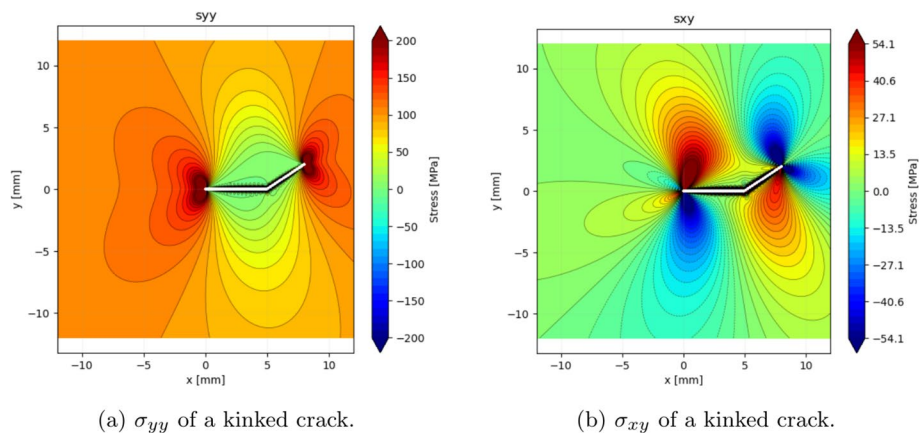


Fig. 9 The stress field around a crack kink. Note the absence of the singular field at the kink (deg-2 vertex), and the strong singularity at both crack tips (deg-1 vertices)

behavior is confined to crack tips, while junctions act as conservative redistribution points within the compatible subspace.

Finite domains

Crack problems in linear elasticity admit a natural boundary-integral representation based on Kelvin fundamental solutions, which satisfy the Navier equations in the interior and reduce the problem to unknown traces on the boundary. The boundary element method (BEM) exploits this structure by recasting the domain problem as an integral equation posed on the outer boundary Γ and the crack surfaces Γ_c (Brebbia 1978; Aliabadi 2002). For fracture problems, this is particularly attractive because crack-induced fields are already available in closed form in infinite media, so boundary effects may be treated as a correction through boundary integral operators. In this sense, coupling VCEM to BEM is structurally natural, since both are based on Green's functions and nonlocal kernel representations.

Relative to finite element discretizations, BEM offers several advantages for linear elastic crack networks. Only the boundary must be discretized, reducing the dimensionality of the problem; crack-tip singular behavior is embedded analytically in the fundamental solution, avoiding special enrichment or singular elements (Belytschko et al. 2009; Aliabadi 1997); and infinite or semi-infinite domains can be treated naturally. Moreover, because VCEM already evaluates crack interactions through dense kernel operators, the addition of dense BEM boundary blocks is algorithmically consistent and fits naturally into the coupled KKT framework. While FEM remains preferable for strongly heterogeneous or nonlinear media, BEM provides an efficient and accurate outer-domain representation for the present linear-elastic setting (Brebbia 1978; Aliabadi 2002).

We consider here two-dimensional isotropic linear elasticity in a bounded domain $\Omega \subset \mathbb{R}^2$ with outer boundary Γ and embedded crack network Γ_c . The boundary fields are displacement and traction: $\mathbf{u} = [u_x, u_y]^T$, $\mathbf{t} = \boldsymbol{\sigma} \mathbf{n}$, where $\mathbf{n}$ denotes the outward unit normal. Let $\mathbf{U}(\boldsymbol{\xi}, \mathbf{x})$ denote the Kelvin displacement tensor, i.e., the displacement at $\boldsymbol{\xi}$ due to a unit point force applied at $\mathbf{x}$ (Eq. 22). The associated traction tensor is given by Equation 24. Applying Betti's reciprocal theorem between the actual field $(\mathbf{u}, \mathbf{t})$ and the Kelvin fundamental solution, and using the divergence theorem, yields the representation formula for any interior point $\boldsymbol{\xi} \in \Omega$:

$$u_j(\boldsymbol{\xi}) = \int_{\Gamma} U_{ji}(\boldsymbol{\xi}, \mathbf{x}) t_i(\mathbf{x}) d\Gamma(\mathbf{x}) - \int_{\Gamma} T_{ji}(\boldsymbol{\xi}, \mathbf{x}; \mathbf{n}) u_i(\mathbf{x}) d\Gamma(\mathbf{x}), \quad (77)$$

which is the Somigliana representation formula for any interior point $\boldsymbol{\xi} \in \Omega$. Given boundary fields $\mathbf{t}$ and $\mathbf{u}$ on Γ , the corresponding elastic layer potentials are defined as

$$\mathcal{S}[\mathbf{t}](\boldsymbol{\xi}) = \int_{\Gamma} \mathbf{U}(\boldsymbol{\xi}, \mathbf{x}) \mathbf{t}(\mathbf{x}) d\Gamma(\mathbf{x}), \quad (78)$$

$$\mathcal{D}[\mathbf{u}](\boldsymbol{\xi}) = \int_{\Gamma} \mathbf{T}(\boldsymbol{\xi}, \mathbf{x}; \mathbf{n}(\mathbf{x})) \mathbf{u}(\mathbf{x}) d\Gamma(\mathbf{x}). \quad (79)$$

The single-layer potential $\mathcal{S}$ represents the displacement field generated by a distribution of boundary tractions acting as force densities. It is weakly singular ($\sim \ln r$ in 2D) and continuous across Γ . The double-layer potential represents a distribution of elastic

dipoles along the boundary. Its kernel is the traction of the Kelvin fundamental solution, which corresponds to the normal derivative of the displacement Green's function. As in classical potential theory, the double-layer potential produces a jump in displacement across the boundary, while its traction remains continuous. This dipole character is responsible for the appearance of the coefficient $c(\boldsymbol{\xi})$ in the boundary-limit form of the Somigliana identity. Its kernel behaves as $O(1/r)$ and exhibits a jump in the limiting value as the observation point approaches Γ from either side.

$$u(\boldsymbol{\xi}) = \mathcal{S}[t](\boldsymbol{\xi}) - \mathcal{D}[u](\boldsymbol{\xi}). \quad (80)$$

As the observation point $\boldsymbol{\xi}$ approaches a smooth point on Γ , the single-layer potential remains continuous. However, the double-layer potential exhibits a jump. Classical potential theory yields

$$\lim_{\boldsymbol{\xi} \rightarrow \Gamma^\pm} \mathcal{D}[u](\boldsymbol{\xi}) = \mp \frac{1}{2} u(\boldsymbol{\xi}) + \text{p.v.} \int_{\Gamma} T(\boldsymbol{\xi}, \mathbf{x}; \mathbf{n}) u(\mathbf{x}) d\Gamma(\mathbf{x}), \quad (81)$$

Substituting this limit into (80) gives the boundary integral equation

$$c(\boldsymbol{\xi}) u(\boldsymbol{\xi}) + \int_{\Gamma} T(\boldsymbol{\xi}, \mathbf{x}; \mathbf{n}) u(\mathbf{x}) d\Gamma(\mathbf{x}) = \int_{\Gamma} U(\boldsymbol{\xi}, \mathbf{x}) t(\mathbf{x}) d\Gamma(\mathbf{x}), \quad (82)$$

where

$$c(\boldsymbol{\xi}) = \begin{cases} \frac{1}{2}, & \text{for smooth boundary points,} \\ \frac{\theta}{2\pi}, & \text{for corners with interior angle } \theta. \end{cases}$$

The outer boundary Γ is partitioned into n_e boundary elements Γ_j , each associated with a collocation point $\boldsymbol{\xi}_i$. Assuming piecewise-constant boundary fields (displacement and traction constant over each element), the boundary integral equation (82) reduces to the discrete system

$$\sum_{j=1}^{n_e} (H_{ij} u_j - G_{ij} t_j) = 0, \quad i = 1, \dots, n_e, \quad (83)$$

with element matrices

$$H_{ij} = \int_{\Gamma_j} T(\boldsymbol{\xi}_i, \mathbf{x}; \mathbf{n}) d\Gamma(\mathbf{x}), \quad (84)$$

$$G_{ij} = \int_{\Gamma_j} U(\boldsymbol{\xi}_i, \mathbf{x}) d\Gamma(\mathbf{x}). \quad (85)$$

Stacking all element values gives the block matrix form

$$\mathbf{H} \mathbf{u}_{\Gamma} - \mathbf{G} \mathbf{t}_{\Gamma} = 0, \quad (86)$$

where

$$\mathbf{u}_{\Gamma} = \begin{bmatrix} u_1 \\ \vdots \\ u_{n_e} \end{bmatrix}, \quad \mathbf{t}_{\Gamma} = \begin{bmatrix} t_1 \\ \vdots \\ t_{n_e} \end{bmatrix}.$$

Let $\Gamma = \Gamma_u \cup \Gamma_t$, where $u = \bar{u}$ is prescribed on Γ_u and $t = \bar{t}$ is prescribed on Γ_t . The complementary traces are unknown. Define the vector of unknown boundary traces

$$y = \begin{bmatrix} u^{bc} \\ t^{bc} \end{bmatrix}, \quad (87)$$

where u^{bc} = unknown displacements on Γ_t , and t^{bc} = unknown tractions on Γ_u . After standard unknown swapping in (86), the BEM system becomes

$$C_{\text{bem}} y = r_{\text{bem}}, \quad (88)$$

where C_{bem} is assembled from sub-blocks of H and G according to the boundary conditions. Thus, the BEM problem solves for the unknown boundary traces y that ensure global equilibrium of the bounded domain.

The crack network introduces additional internal degrees of freedom $q \in \mathbb{R}^{n_q}$ (Burgers density DOFs). These produce displacement and traction traces on the outer boundary via the infinite-medium crack kernels:

$$u_{\Gamma}^{cr} = M_u q, \quad t_{\Gamma}^{cr} = M_t q. \quad (89)$$

Hence, the total boundary traces decompose as

$$u_{\Gamma} = u^{bc} + M_u q, \quad t_{\Gamma} = t^{bc} + M_t q. \quad (90)$$

Substituting into (86) yields the coupled system

$$H(u^{bc} + M_u q) - G(t^{bc} + M_t q) = 0. \quad (91)$$

Rearranging,

$$C_{\text{bem}} y + Bq = 0, \quad (92)$$

where

$$B = HM_u - GM_t.$$

Thus, y enforces global boundary equilibrium, q represents internal crack degrees of freedom, and Bq represents the boundary disturbance induced by cracks. In the infinite-domain formulation, the crack DOFs q are determined by minimizing the elastic energy subject to admissibility constraints:

$$H_{cr} q + C^T \lambda = f_{cr}.$$

In the bounded-domain case, the crack DOFs remain governed by the same energy functional and KKT constraints. However, the external traction vector f_{cr} is no longer prescribed directly; it is influenced by the bounded domain through the BEM system.

The coupled problem, therefore, has the structure

$$\begin{bmatrix} H_{cr} & C^T & B^T \\ C & 0 & 0 \\ B & 0 & C_{\text{bem}} \end{bmatrix} \begin{bmatrix} q \\ \lambda \\ y \end{bmatrix} = \begin{bmatrix} f_{cr} \\ 0 \\ 0 \end{bmatrix}.$$

This structure is fully consistent with the infinite-medium KKT formulation: the crack block remains variational and saddle-point in nature, while the BEM block enforces

global boundary equilibrium. The outer boundary, therefore, acts as an additional constraint coupled to the crack-induced fields.

The crack-network equilibrium, including boundary-induced loading, is written in algebraic form as

$$K_{qq}q + Ny = f_{vc}. \quad (93)$$

Here: $y = [u^{bc}; t^{bc}]$ are boundary correction DOFs, N maps boundary corrections to tractions acting on cracks, f_{vc} is the applied external loading on the cracks. The second term Ny represents the feedback of the bounded elastic domain onto the crack network. The BEM self consistency implies that $C_{bem}y = 0$, while the outer boundary conditions with crack traces require that $C_{bc}y + Mq = d$. The matrix C_{bem} enforces the discrete boundary integral equation (self-equilibrium of the bounded elastic domain), while C_{bc} enforces prescribed boundary conditions. The term Mq represents crack-induced traces on the outer boundary.

Crack side algebraic constraints are enforced as: $C_{vc}q = g_{vc}$. The resulting augmented KKT system is thus:

$$\begin{bmatrix} K_{qq} & N & C_{vc}^T & M^T \\ 0 & C_{bem} & 0 & C_{bc}^T \\ C_{vc} & 0 & 0 & 0 \\ M & C_{bc} & 0 & 0 \end{bmatrix} \begin{bmatrix} q \\ y \\ \lambda \\ \mu \end{bmatrix} = \begin{bmatrix} f_{vc} \\ 0 \\ g_{vc} \\ d \end{bmatrix}. \quad (94)$$

The resulting system is generally non-symmetric because the BEM block C_{bem} is derived from a collocation discretization of the strong-form boundary integral equation, whereas the crack block K_{qq} arises from an energy-based variational formulation.

Three operational modes are currently implemented in the VCEM code, depending on the application and required accuracy:

1. **No coupling:** The stress field due to the external boundary conditions is computed once using standalone BEM, and the resulting traction field drives the crack solve. Crack-induced boundary perturbations are neglected.
2. **Iterative coupling:** Crack-induced boundary tractions are applied, together with the external boundary conditions, to the BEM model, and the updated boundary field is used to drive the next crack solve. This procedure is iterated until convergence.
3. **Direct (monolithic) coupling:** The full augmented KKT system (94) is solved simultaneously for (q, y, λ, μ) . This can be an expensive computation if applied to every crack growth increment.

In all modes, coupling updates are applied only at *outer growth cycles* to limit computational cost. A block diagram illustrating the key ingredients in the enhanced BEM-KKT formulation is shown in Fig. 10

Before closing this section, we comment of a few numerical aspects of the BEM-KKT system, since the monolithic system is indefinite and sensitive to scaling. Practical issues involve conditioning of dense BEM operators, mixed-unit disparity between crack and boundary DOFs, and weakly observable crack modes in coupling maps. For these reasons, we use a traction non dimensionalization in the implementation

$$\tilde{t} = \frac{L}{G}t, \quad \tilde{M}_t = \frac{L}{G}M_t, \quad \tilde{\bar{t}} = \frac{L}{G}\bar{t},$$

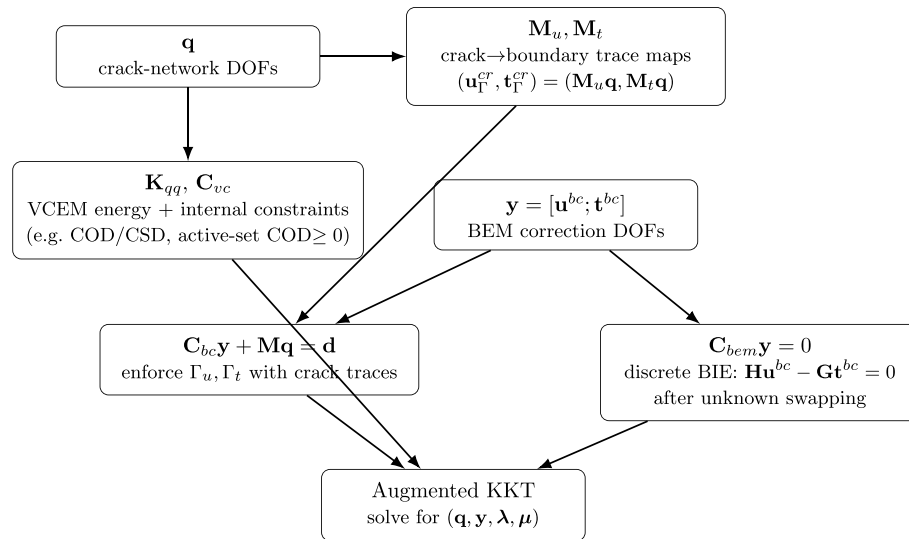


Fig. 10 Block diagram of the coupled BEM-KKT system solution method

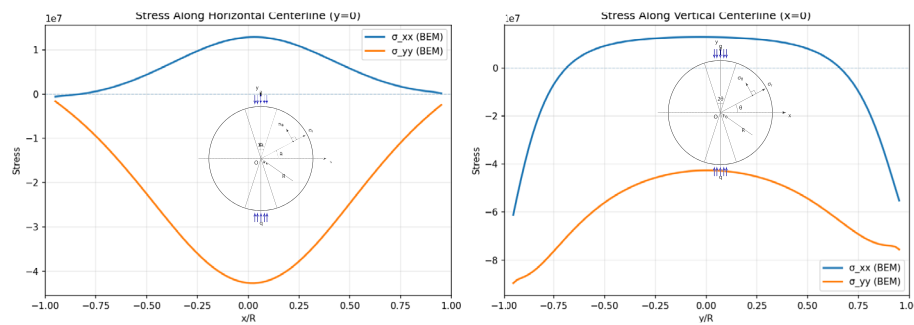


Fig. 11 Stresses σ_{xx} and σ_{yy} along the vertical and horizontal centerlines of the compressed disk

with characteristic length L taken as the boundary diameter. Additional row/variable equilibration and mild ridge regularization are used for numerical robustness.

Validation of the BEM block

The numerical method was validated first against the analytical solution of a disk compressed on two opposing sides by radial pressures applied over a small arc. This is a classic problem for which analytical results are available and can be used to benchmark the accuracy of the solution. The parameters used in disk compression validation are as follows: diameter of 25.4 mm, total applied force of 3.8 kN, $E = 231.52$ GPa, $\nu = 0.3$, number of boundary elements = 120, pressure arc angle $= 30^\circ$ and disk thickness of 6.35 mm. These parameters correspond to our experiment with alumina (Alabdullah et al. 2023).

Figures 11 and 12 validate the outer-boundary BEM block by comparing the calculated stress field in a circular disk subjected to opposing radial pressures applied over two short arc patches to available closed-form solutions for the same boundary-value problem. The benchmark uses the experimental geometry and loading reported earlier. The imposed pressure is obtained from force equilibrium over the discrete lengths of the loaded arc (about 1.6 mm per patch at unit thickness, corresponding to 2.38 MPa).

Figure 11 shows σ_{xx} and σ_{yy} along the horizontal and vertical centerlines. The qualitative structure follows from symmetry and the nature of diametral compression: the

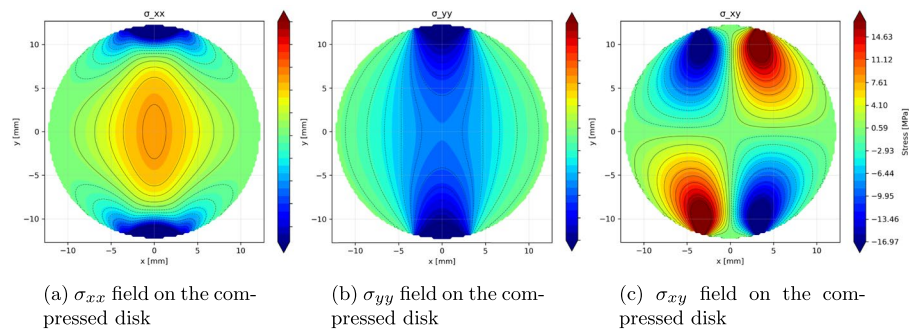


Fig. 12 Stress component distributions in the compressed disk under load control

dominant component σ_{yy} is compressive along the loaded vertical diameter, while σ_{xx} develops as a tensile field throughout the interior along the horizontal diameter. This interior tensile field is the classical mechanism underlying splitting in the Brazilian test: despite compressive loading at the boundary, the induced stress state contains a tensile component transverse to the loading axis that is largest near the disk center in the idealized elastic solution (Hondros 1959). Quantitatively, the BEM solution reproduces the analytical center stresses to within a few percent. For the present parameters, the center values computed are $\sigma_{xx} = 9.09 \times 10^4$ Pa and $\sigma_{yy} = -2.80 \times 10^5$ Pa (with σ_{xy} essentially zero), with relative errors reported in the center of approximately -4.52% in σ_{xx} and -2.04% in σ_{yy} compared to the analytical solution (Gontarz and Podgórski 2017; Hondros 1959).

Figure 12 shows the spatial distributions of σ_{xx} , σ_{yy} , and σ_{xy} over the disk. The σ_{yy} field is dominated by compressive regions beneath the loaded arcs and along the loading line, while σ_{xx} exhibits an interior tensile region spanning the mid-plane normal to the applied load; this is the same structure responsible for diametral splitting in the analytical Brazilian-disk solution (Hondros 1959). The shear stress σ_{xy} displays the expected antisymmetric lobes in the quadrants and vanishes on the principal symmetry axes (centerlines) by the symmetry of both geometry and loading. The close agreement of (i) centerline profiles and (ii) full-field contour structure with analytical expectations supports the correctness of the BEM operator assembly, the handling of mixed boundary data on the loaded/unloaded patches, and the numerical quadrature strategy for near-singular panel interactions. This benchmark, therefore, provides a reliable outer-boundary solver component for subsequent crack-network simulations in finite domains.

Figure 13 shows the stress distributions when the disk is loaded under displacement control, implemented by prescribing a vertical downward displacement of $-4 \mu\text{m}$ over the top 15° arc, while constraining the bottom arc to $y = 0$ to mimic platen contact. Although the qualitative structure remains consistent with diametral compression, important differences arise relative to the load-controlled case of Fig. 12. Under load control, the imposed pressure produces a stress field whose magnitude is directly proportional to the applied traction, leading to a pronounced interior tensile σ_{xx} band along the horizontal diameter and strong compressive σ_{yy} beneath the loaded arcs. In contrast, under displacement control, the reaction forces are not prescribed a priori but emerge from the compatibility constraint between the platen and the disk; consequently, the stress field redistributes in a manner that reflects global stiffness rather than local traction magnitude. The compressive σ_{yy} lobes beneath the contact arcs become more

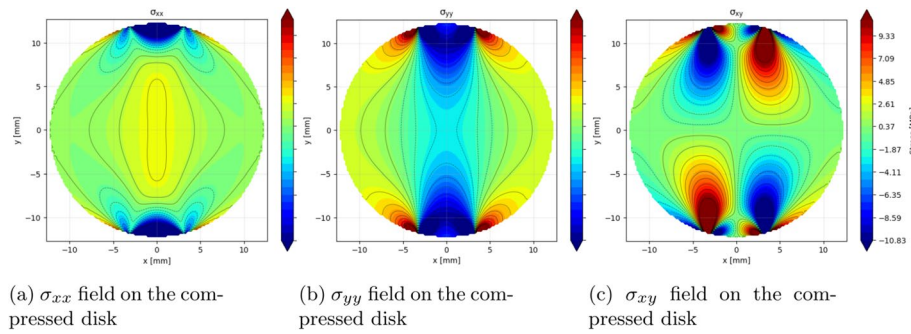


Fig. 13 Stress component distributions in the compressed disk under displacement control

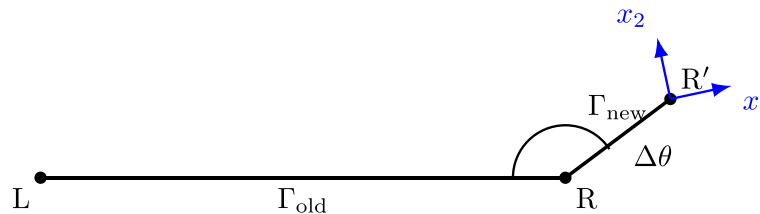


Fig. 14 Single-tip extension: append a new segment at the active (right) tip. Old DOFs remain; new DOFs are added only on the appended segment

sharply localized, while the interior tensile σ_{xx} region tends to flatten and redistribute, reflecting the fact that the system adjusts its reaction forces to maintain the imposed displacement.

From the standpoint of crack evolution, these differences are significant. Under load control, crack extension increases compliance and therefore amplifies the energy release rate, promoting unstable propagation once the critical condition is met. Under displacement control, however, crack growth partially relaxes the reaction forces at the contact arcs, reducing the incremental driving force and tending to stabilize propagation. For a central crack aligned with the horizontal diameter, the interior tensile σ_{xx} field remains the primary opening mechanism in both cases; however, the magnitude and spatial extent of this tensile zone are sensitive to the loading protocol. Displacement control therefore modifies both the peak K_I and the gradient of the energy release rate with respect to crack length. As the crack approaches the boundary, the confinement-induced reduction in opening is expected to occur earlier under displacement control since the reaction forces diminish as compliance increases. These distinctions underscore that the Brazilian disk is not merely a geometric fracture benchmark but also a thermodynamic one: the loading protocol directly influences the stability landscape governing crack advance, which will be discussed in the companion paper (Ghoniem 2026).

Crack network evolution

Once the equilibrium solution has been obtained for the current crack configuration, crack extension is evaluated using local propagation criteria based on the near-tip fields. The present framework, therefore, combines a constrained variational equilibrium solver with a separate criterion-based crack-growth rule. Crack propagation proceeds by extending crack elements with an additional polar crack at its tip, as shown in Fig. 14. The direction, driving force, and admissibility of such growth are governed by fracture

criteria derived from continuum fracture mechanics, augmented when needed by stochastic or microstructural considerations. Crack branching is not posed as a fully general variational topological transition but is introduced through additional geometric update rules based on the local crack-tip state. Likewise, irreversibility is not embedded here as a full history-dependent inequality constraint within the variational problem itself. Rather, it is enforced only in the geometric sense that crack updates are additive: new crack segments may be introduced, but previously created crack surfaces are not removed or healed.

The crack-growth increment Δa is treated here as a numerical step-size parameter. In practice, it should be chosen to be small relative to the local crack length scale and the discretization near the active tip, so that the near-tip fields used to determine the propagation direction remain locally representative. After each extension step, the crack polyline is updated geometrically and then re-parameterized/re-discretized before solving the next equilibrium problem. The overall algorithm therefore proceeds as a sequence of equilibrium solves on successively updated crack geometries. As in other local crack-growth methods, step-size sensitivity should be checked, since excessively large Δa may bias the computed crack path, whereas smaller increments improve path convergence at increased cost.

In this section, we summarize (i) criteria for determining the crack growth direction, (ii) methods for computing stress intensity factors and energy release rates within the present formulation, and (iii) the propagation criteria based on deterministic thresholds or stochastic activation.

Determination of the stress intensity factors

Williams Expansion Using Williams expansion of the displacement field (Williams 1956), the primary displacement-based estimator fits opening/sliding jumps to

$$\text{COD}(r) \approx A_I \phi(r) + B_I \phi(r) \bar{r}, \quad \text{CSD}(r) \approx A_{II} \phi(r) + B_{II} \phi(r) \bar{r}, \quad (95)$$

with $\phi(r) = \sqrt{r/(2\Phi)}$ and $\bar{r} = r/(2\Phi)$. The fitted amplitudes are converted through

$$K_I = \frac{\mu}{\kappa + 1} A_I, \quad K_{II} = \frac{\mu}{\kappa + 1} A_{II}, \quad (96)$$

with κ as the Kolosov constant. The distance, r is taken as the edge distance $a \mp x$ or the Euclidean tip distance $\|\mathbf{x} - \mathbf{x}_{\text{tip}}\|$. This is usually the default estimator for the SIF in the VCEM code. It is inexpensive, capable of mixed-mode, and robust when the fit window is chosen in the asymptotic zone. It is best for routine static post-processing on both straight and branched geometries.

A reduced estimator regresses jump values against $\sqrt{r}$. If $\Delta u(r) \approx A\sqrt{r}$, then

$$K \approx \frac{E' \sqrt{2\Phi}}{8} A, \quad (97)$$

with $E' = E$ (plane stress) and $E' = E/(1 - \nu^2)$ (plane strain). Applied separately to COD and CSD, this gives K_I and K_{II} . Its merits are speed and simplicity. It is most useful for rapid screening and cross-checking but is less stable than the two-term Williams fit in noisy or strongly curved near-tip fields.

Peach-Koehler Force In the absence of explicit dislocation motion, the Peach–Koehler force is energetically defined as the configurational force associated with a virtual advance of the crack tip. A local configurational force is accumulated over a near-tip panel window,

$$F_{PK} \approx \sum_{k \in \mathcal{W}} ([\sigma(\mathbf{x}_k) d\mathbf{B}_k] \times \mathbf{e}_z), \quad d\mathbf{B}_k = (b_{II,k} \mathbf{t}_k + b_{I,k} \mathbf{n}_k) \Delta s_k, \quad (98)$$

then projected as $J_1 = F_{PK} \cdot \mathbf{t}$, $J_2 = F_{PK} \cdot \mathbf{n}$. The conversion uses

$$J_1 = \frac{K_I^2 + K_{II}^2}{E'}, \quad J_2 = -\frac{2K_I K_{II}}{E'}, \quad (99)$$

with inversion

$$A = E' J_1, \quad C = -\frac{1}{2} E' J_2, \quad D = \sqrt{A^2 - 4C^2}, \quad (100)$$

$$K_I = \sqrt{\frac{1}{2}(A + D)}, \quad K_{II} = \text{sign}(C) \sqrt{\frac{1}{2}(A - D)}. \quad (101)$$

This estimator is advantageous when growth is interpreted through configurational driving force and when coupling to direction criteria is needed. It is most reliable when stress reconstruction is accurate and self-panel contamination is excluded. In some cases (like the arc crack) the SIF components must be rotated from a local frame to a global frame, as shown in Eq. (102) (Erdogan and Sih 1963; Cotterell and Rice 1980).

$$\begin{aligned} K'_I &= K_I \cos^3\left(\frac{\theta}{2}\right) - \frac{3}{2} K_{II} \sin\left(\frac{\theta}{2}\right) \cos^2\left(\frac{\theta}{2}\right), \\ K'_{II} &= K_I \sin\left(\frac{\theta}{2}\right) \cos^2\left(\frac{\theta}{2}\right) + K_{II} \cos\left(\frac{\theta}{2}\right) \left(1 - 3 \sin^2\left(\frac{\theta}{2}\right)\right). \end{aligned} \quad (102)$$

Crack growth direction

Consider a crack tip located at $\mathbf{X}_{\text{tip}}$ with local coordinate system (x_1, x_2) aligned with the current crack segment. A candidate extension is introduced as a new half-crack making an angle θ with respect to the local x_1 axis. Several classical criteria are commonly employed to determine θ :

Maximum hoop stress criterion (MTS) The crack propagates in the direction that maximizes the circumferential (hoop) stress $\sigma_{\theta\theta}$ ahead of the tip. For mixed-mode loading (K_I, K_{II}) , this yields the condition

$$\frac{\partial \sigma_{\theta\theta}(r, \theta)}{\partial \theta} = 0, \quad (103)$$

which leads to the well-known analytical solution (Erdogan and Sih 1963):

$$\theta_{\text{MTS}} = 2 \arctan \left[\frac{1}{4} \left(\frac{K_I}{K_{II}} \pm \sqrt{\left(\frac{K_I}{K_{II}} \right)^2 + 8} \right) \right] \quad (104)$$

This criterion is widely used due to its simplicity and robustness.

Minimum strain energy density criterion (SED). The crack grows in the direction that minimizes the strain energy density factor $S(\theta)$ near the tip. The preferred direction satisfies (Sih 1974)

$$\frac{\partial S(\theta)}{\partial \theta} = 0, \quad (105)$$

where $S(\theta)$ is expressed in terms of K_I , K_{II} , and the elastic constants. This criterion is particularly attractive when both modes contribute significantly.

Local symmetry criterion. Crack growth is assumed to occur in a direction such that the shear mode vanishes at the new tip: $K_{II}(\theta) = 0$. This condition enforces symmetry of the near-tip stress field and is naturally compatible with dislocation-based formulations.

Geometric and obstacle-aware criteria. In heterogeneous materials, the growth direction may be constrained to avoid obstacles such as inclusions, grain boundaries, or pre-existing defects. In such cases, θ is chosen to minimize a cost functional incorporating both fracture driving force and geometric constraints.

Propagation criteria

Once the direction and driving force are known, crack advance is determined by a propagation criterion.

Deterministic threshold criteria. The simplest criterion assumes that crack growth occurs when a critical value is exceeded:

$$K_{\text{eq}} \geq K_{IC}, \quad \text{or equivalently} \quad G \geq G_c, \quad (106)$$

where K_{eq} is an appropriate equivalent stress intensity factor (e.g. $\sqrt{K_I^2 + K_{II}^2}$), and K_{IC} or G_c are material fracture toughness parameters.

Rate-dependent and dynamic criteria. For dynamic fracture, propagation may depend on the crack-tip velocity and inertia. In this case, G and K are evaluated using dynamic (retarded) kernels, and crack advance may be governed by a relation of the form: $\dot{a} = \mathcal{F}(G - G_c)$, where $\dot{a}$ is the crack growth rate.

Stochastic and microstructure-aware criteria. In materials with distributed microscopic flaws, crack advancement may be treated probabilistically. Local fracture resistance is modeled as a random field, and propagation occurs when $G(X_{\text{tip}}) > G_c(X_{\text{tip}})$, with G_c sampled from a prescribed statistical distribution. This approach naturally captures scatter, crack deflection, and intermittent growth. An equivalent driving stress intensity factor is defined as

$$K_{\text{eff}} = \sqrt{K_I^2 + K_{II}^2}. \quad (107)$$

Let the fracture toughness be a spatially varying field, $K_c = K_c(x, y)$. For a crack tip located at (x_t, y_t) , propagation is permitted if $K_{\text{eff}}(x_t, y_t) > K_c(x_t, y_t)$. If this condition is not satisfied, the tip remains stationary during the current growth step. For propagating tips, the crack growth angle θ is computed relative to the local crack tangent using

a mixed-mode fracture criterion. In general, $\theta = \Theta(K_I, K_{II})$, where Θ may represent, for example, the maximum hoop stress criterion. The absolute propagation direction is then $\phi = \phi_{\text{tangent}} + \theta$. For cracks with multiple propagating tips, propagation may be handled simultaneously for all active tips or sequentially, with the choice fixed consistently throughout the simulation. Simultaneous propagation avoids ordering bias and is preferred when computationally feasible. The adaptive method of crack propagation can be computationally expensive, especially for large crack networks. In practice, however, a fixed crack increment is used based on trial simulations to establish the appropriate growth step size.

The formulation can accommodate curved cracks, but its performance for highly curved paths depends on adequate local geometric resolution. When the crack exhibits a tight bend or a small radius of curvature, the discretization must be sufficiently refined to resolve the local change in tangent direction and the associated near-field variation. If the curvature is under-resolved, errors may appear in the local stress field, stress-intensity evaluation, and hence in the predicted propagation direction. For this reason, highly curved cracks are a regime in which local refinement and curvature-aware discretization become especially important.

Conclusions and outlook

We established here the Variational Crack Element Method (VCEM) as a topology-preserving, constrained variational framework for sharp-interface fracture mechanics. By representing cracks through distributed Burgers densities and expressing their interactions via elastostatic dislocation kernels, the method embeds the exact LEFM structure directly into the approximation space through basis functions. The resulting formulation retains the classical sharp crack geometry of linear elastic fracture mechanics while recasting the governing equations as a constrained energy-minimization problem. In contrast to purely strong-form boundary integral approaches, the present method derives equilibrium as the Euler–Lagrange condition of an elastic energy functional, thereby enabling the systematic incorporation of junction compatibility, Burgers-vector conservation, and unilateral contact constraints within a unified KKT structure. The algebraic form of the governing equations is independent of crack geometry complexity; branching, kinking, and network coalescence modify only the constraint operators and graph connectivity, not the underlying variational principle. In this sense, VCEM closes a long-standing gap between classical distributed dislocation techniques, boundary element elasticity, and modern constrained optimization theory, providing a mathematically transparent and computationally modular foundation for interacting crack networks in bounded or unbounded domains. While we focused here on the mathematical formulation and algorithmic implementation of VCEM, in the companion paper, we show that the method is validated for a high degree of accuracy against available analytical and complex variable boundary integral solutions in a companion paper (Ghoniem 2026).

The present formulation relies on the availability of explicit fundamental solutions for the governing elasticity problem, here the Kelvin solution for 2D linear isotropic elasticity. This is the source of both its efficiency and its sharp-crack fidelity, but it also defines its current scope. In particular, direct extension to nonlinear materials, strongly heterogeneous media, or large-deformation settings is not immediate, since the corresponding

Green's functions may be unavailable or may not admit a useful superposable representation. More general extensions would require replacing the analytical interaction kernels with numerically constructed influence operators or coupling the present crack-network formulation to an outer numerical solver that resolves the nonlinearity or heterogeneity at the continuum level.

The formulation presented here suggests a broader perspective on fracture mechanics as a constrained variational field theory on evolving graph manifolds. Because the governing operators are kernel-based and translation-invariant in global coordinates, they are inherently compatible with hierarchical compression and fast summation strategies, enabling near-linear complexity for large networks. The saddle-point structure that arises from equality and inequality constraints is well aligned with modern large-scale linear algebra, including Krylov subspace solvers, block and physics-based preconditioning, null-space projections, and adaptive active-set methods for contact. These properties have the potential to open the path toward simulations involving 10^4 – 10^5 crack degrees of freedom while preserving variational consistency and topological fidelity. Beyond static fracture, the extension to dynamic problems follows directly by replacing static Green's functions with retarded elastodynamic kernels, maintaining the same constrained structure. The kernel-dominated algebra also maps naturally onto contemporary high-performance computing architectures: dense kernel interactions and batched tensor contractions are ideally suited to GPU acceleration, while graph-based crack topology management integrates naturally within distributed-memory (MPI) environments. At a conceptual level, the framework provides a bridge between discrete crack mechanics, network percolation phenomena, and variational principles of energy release, suggesting future connections between fracture, phase transition theory, and data-driven materials modeling. In this broader view, VCEM offers not only a numerical method but a structural paradigm for scalable, topology-aware fracture simulation in complex materials and engineering systems.

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Authors' contributions

Nasr M. Ghoniem conceived the research, developed the theoretical formulation, designed and implemented the computational methodology, performed the numerical simulations, analyzed the results, and wrote and revised the manuscript. The author has read and approved the final manuscript.

Data availability

All data supporting the findings of this study are available within the article and its Supplementary Information.

Declarations

Ethics Declaration

The author declares that this work is original, has been conducted in accordance with accepted standards of research integrity, and does not involve human or animal subjects.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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